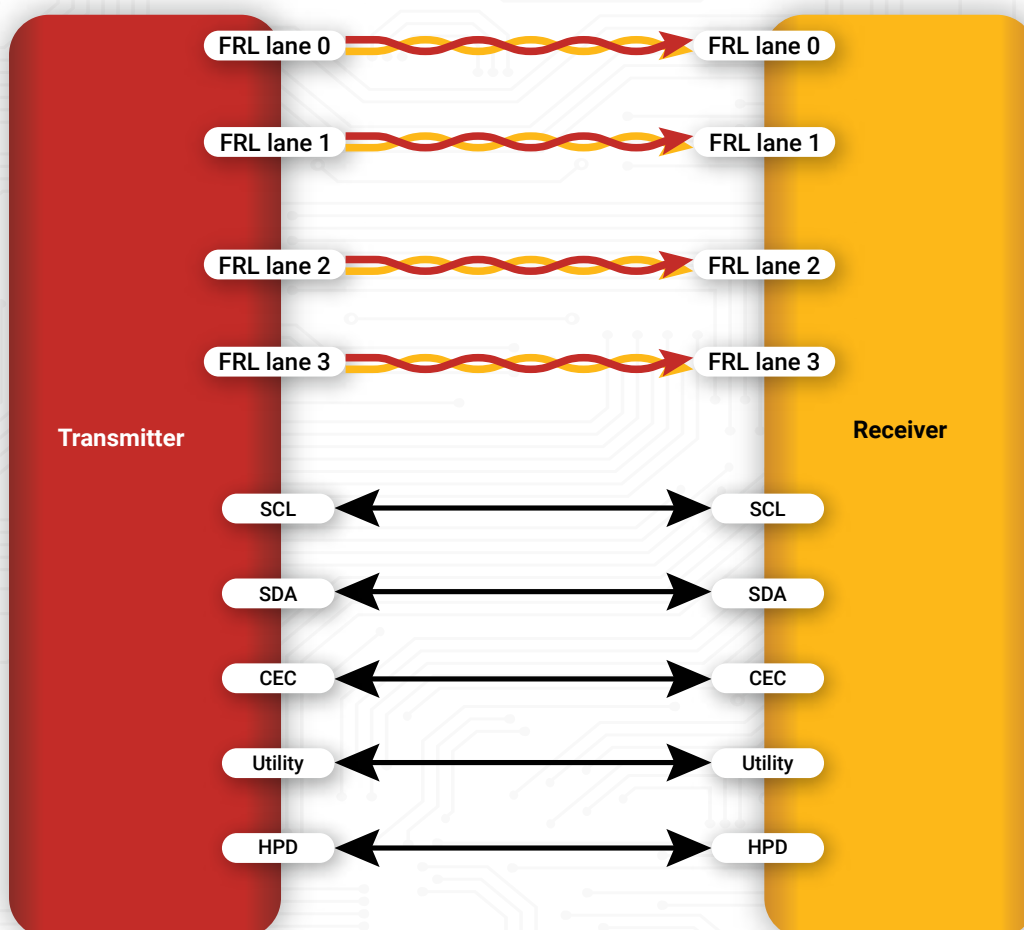


# PCB Interfaces and Communication Protocols Handbook



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## 1. Introduction

The choice of the PCB interface plays a critical role in enabling efficient protocol layering, minimizing crosstalk, and supporting advanced techniques like differential signaling and adaptive equalization.

As a PCB designer, understanding circuit board interfaces will help you choose and implement the right one. This ensures minimal signal loss, compatibility between components, and prevents system failures or board respins.

### Highlights:

1. PCB interfaces are electrical and logical connections that enable data, power, and control signal transmission through protocols like SPI, I2C, Bluetooth, or Wi-Fi.
2. Circuit board interfaces are categorized into wired (e.g., UART, I2C, PCIe) and wireless (e.g., Bluetooth, Wi-Fi, LoRa).
3. The choice of interface depends on factors like data transfer speed, communication distance, system complexity, power consumption, and environmental conditions.

In this eBook, you'll learn about wired and wireless PCB interfaces and their roles in enabling communication between components.

Each interface is briefly explained, followed by a table highlighting its key characteristics, such as communication type, speed, distance, typical applications, and pros and cons.

## 2. What are PCB interfaces?

Printed board interfaces are the electrical and logical connections that enable communication between the board and other components or devices. They define how data, power, or control signals are transmitted using specific protocols or standards, such as SPI, I2C, Bluetooth, or Wi-Fi.

An interface isn't the same as a PCB connector. A connector is the physical part (like a port or plug) that connects the board to external devices. In contrast, an interface is the entire system that works over that physical link, including the signal lines and the protocols.



### 3. What are the different types of circuit board interfaces?

PCB interfaces are categorized based on how they connect and communicate with other devices or components.

They can be broadly divided into two types:

1. **Wired interfaces** (like UART, I2C, or PCIe) use physical connections, such as cables or traces, to transmit data, power, or control signals.
2. **Wireless interfaces** (like Bluetooth, Wi-Fi, or LoRa) transmit data without physical connections, using radio waves or other wireless technologies. These are ideal for remote communication or applications where wiring is impractical.

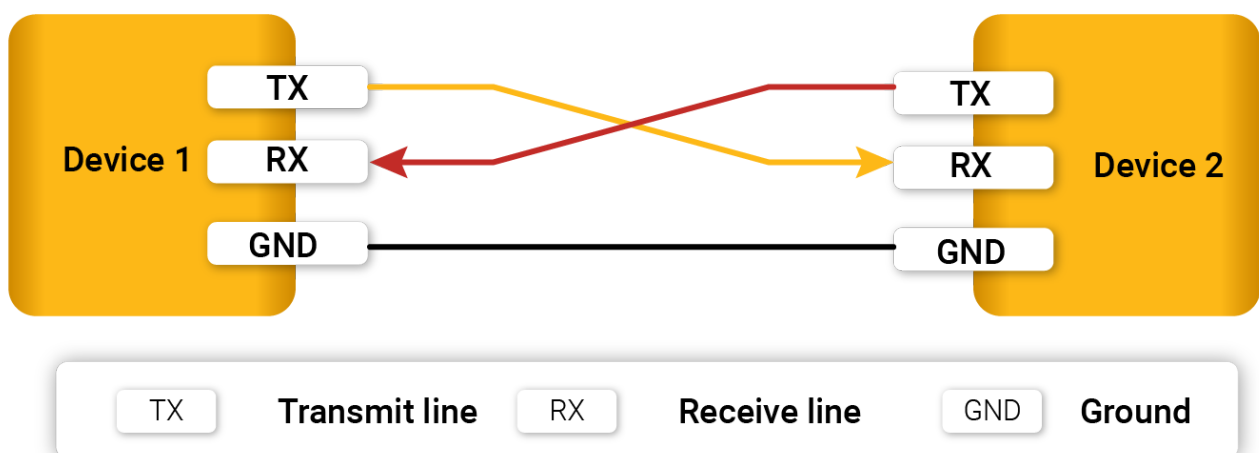
The choice of the interface depends on the signal speed, distance, complexity, and power requirements.

### 4. 19 wired PCB interfaces every designer should know

#### 4.1. Universal asynchronous receiver/transmitter (UART)

UART is a simple asynchronous serial communication protocol that transmits data one bit at a time.

##### UART communication setup



Communication using the UART PCB interface for point-to-point serial data exchange between two devices

Unlike SPI or I2C, UART does not use a clock signal; instead, both devices must agree on a baud rate beforehand.

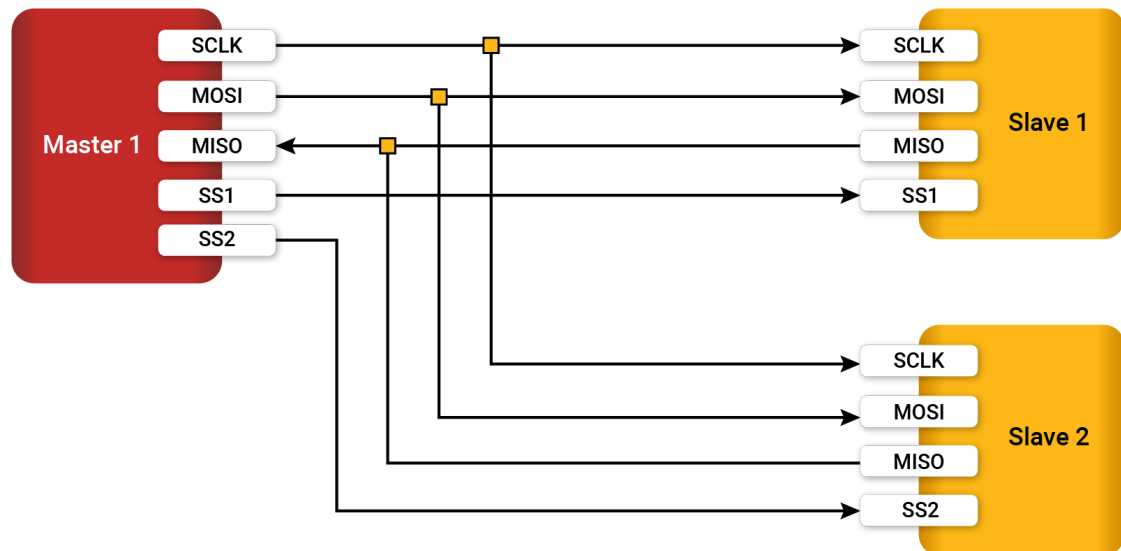
**Table 1:** UART interface features

| Factors            | Details                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| Type               | Asynchronous serial communication                                                                                      |
| Data lines         | Transmit (TX), receive (RX)                                                                                            |
| Communication mode | Full-duplex                                                                                                            |
| Number of devices  | 2 (point-to-point)                                                                                                     |
| Speed              | Typically 230 kbps to 460 kbps Baud rates: 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600, 1000000, 1500000 |
| Distance           | Up to 15 meters with RS-232 with lower baud rate; TTL UART is limited to 1–2 meters.                                   |
| Applications       | Microcontroller to peripheral communications, GPS, Bluetooth modules                                                   |
| Advantages         | Simple design, low resource usage                                                                                      |
| Disadvantages      | Limited distance, basic error detection (parity only), no built-in error correction                                    |

## 4.2 Serial peripheral interface (SPI)

SPI is a widely used interface for high-speed, full-duplex communication between microcontrollers and peripheral ICs like sensors, ADCs, DACs, shift registers, and SRAM. It allows simultaneous data transmission by both master and slave devices.

## SPI network with multiple slave devices



SCLK    Serial clock    MISO    Master-in slave-out  
 MOSI    Master-out-slave-in    SS    Slave-select

Communication between one master and multiple slave devices using the SPI PCB interface

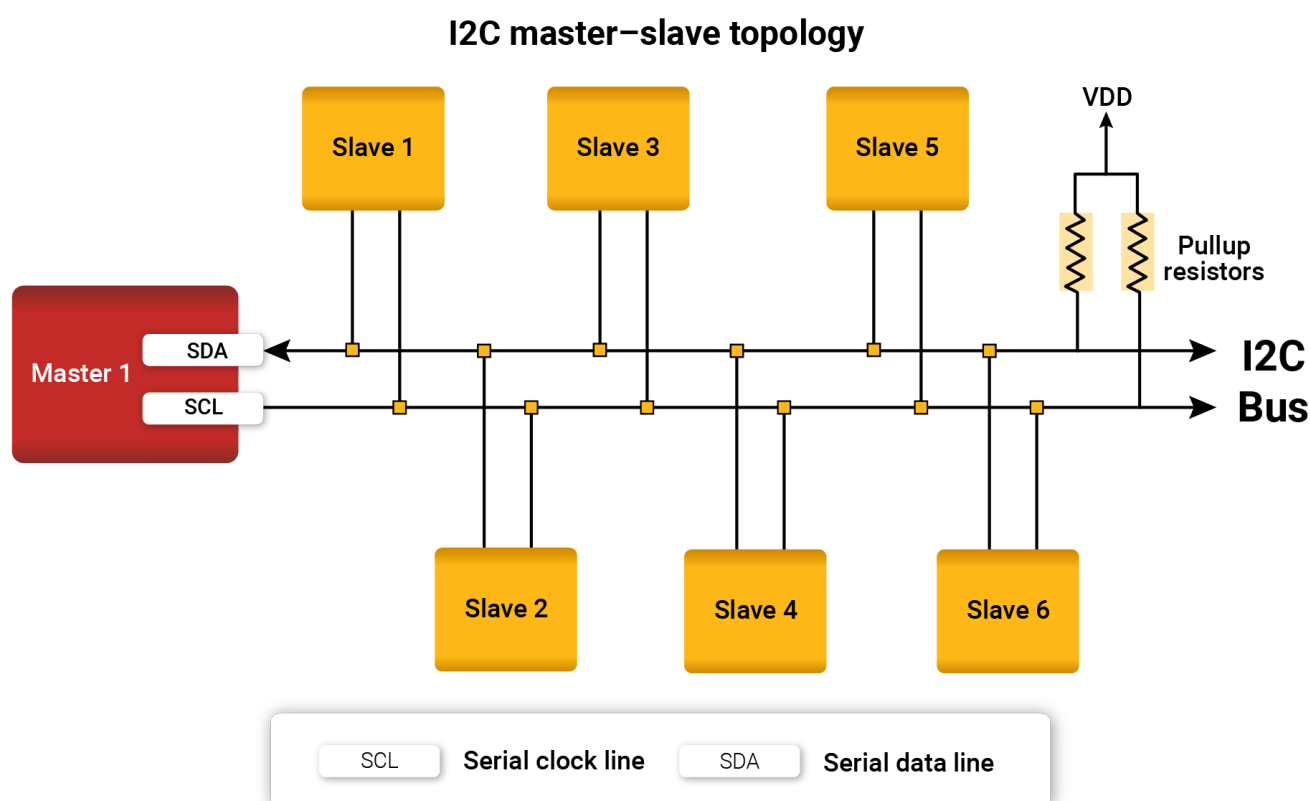
**Table 2: SPI features**

| Factors            | Details                                                              |
|--------------------|----------------------------------------------------------------------|
| Type               | Synchronous serial communication                                     |
| Data lines         | MOSI, MISO, SCLK, SS                                                 |
| Communication mode | Full-duplex                                                          |
| Number of devices  | Typically 1 master, up to 4–5 slaves (more with extra decoder logic) |
| Speed              | Typically, 1 Mbps up to 20 Mbps                                      |
| Distance           | Short distance (PCB-level)                                           |
| Applications       | Sensors, memory chips, and display modules                           |

| Factors       | Details                                                                        |
|---------------|--------------------------------------------------------------------------------|
| Advantages    | High-speed, full-duplex communication                                          |
| Disadvantages | Short distances, needs multiple chip select lines, no standard error detection |

### 4.3 Inter-integrated circuit (I2C)

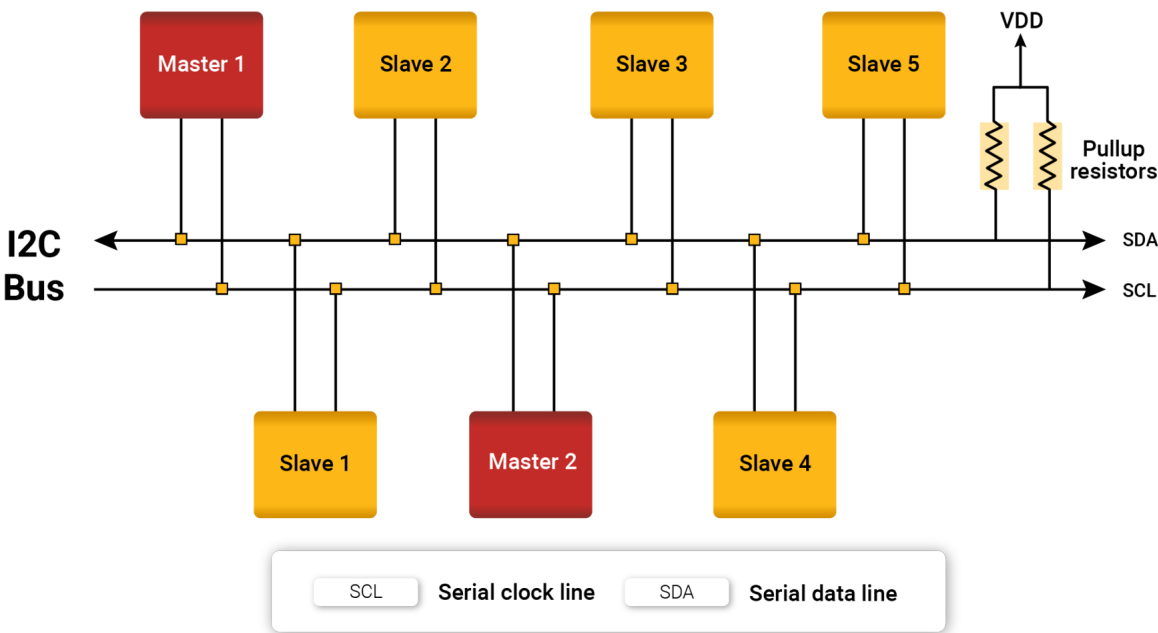
I2C is a two-wire serial communication protocol that allows multiple devices to share a single communication bus. It supports multiple controllers capable of transmitting and receiving data and commands.



**I2C two-wire serial interface for communication between a master and multiple slave devices**

Data is transferred in byte-sized packets, with each target device identified by a unique address.

I2C multi-master architecture using master-slave communication



I2C synchronous serial communication between two master devices and multiple slave devices on a shared bus

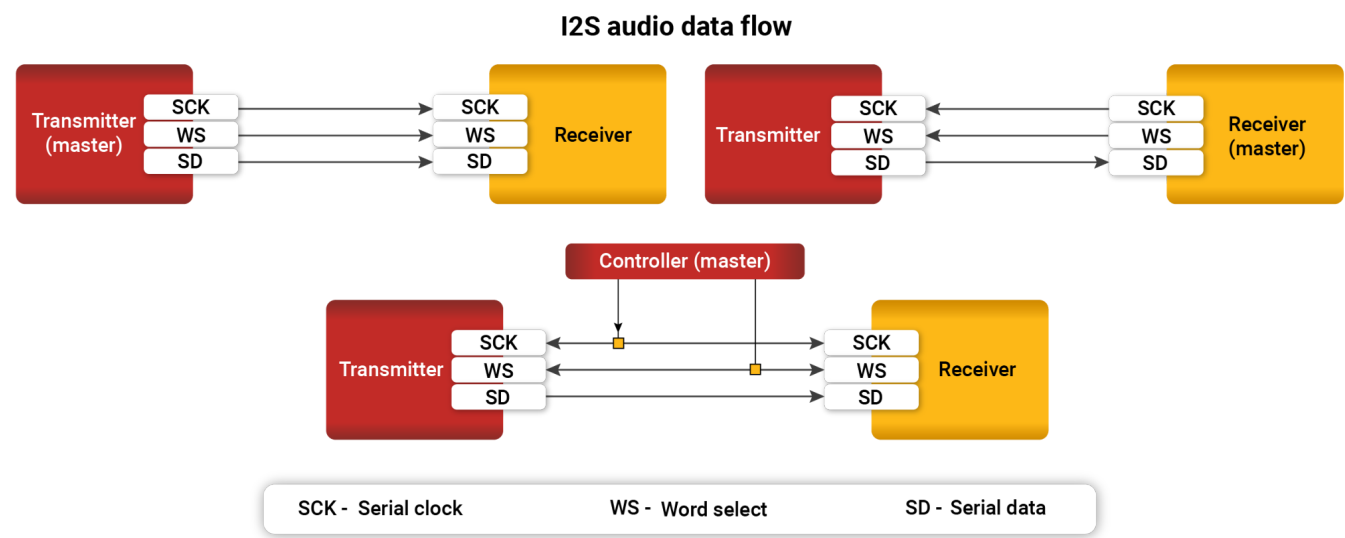
Table 3: I2C interface features

| Factors            | Details                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------|
| Type               | Synchronous serial communication (multi-master, multi-slave)                                                             |
| Data lines         | Data (SDA), clock (SCL)                                                                                                  |
| Communication mode | Half-duplex (shared bus)                                                                                                 |
| Number of devices  | 1 master, up to 127 slave devices                                                                                        |
| Speed              | 100 kbps (standard), 400 kbps (fast-mode), 1 Mbps (fast-mode plus), 3.4 Mbps (high-speed mode), 5 Mbps (ultra-fast mode) |
| Distance           | Short distance (1-2 meters)                                                                                              |
| Applications       | EEPROMs, sensors, and RTC modules                                                                                        |

| Factors       | Details                                          |
|---------------|--------------------------------------------------|
| Advantages    | Only two wires needed, supports multiple devices |
| Disadvantages | Slower than SPI, limited distance                |

### 4.4 Inter-IC sound (I2S)

I2S is a synchronous serial communication protocol designed specifically for transmitting digital audio data between integrated circuits.



I2S PCB interface configurations showing master-slave roles with the transmitter, receiver, and an external controller acting as the clock master

Unlike UART, I2S uses a clock signal to synchronize data transmission and supports separate lines for audio data, word select, and bit clock, making it ideal for high-quality stereo audio streaming.

**Table 4:** I2S interface features

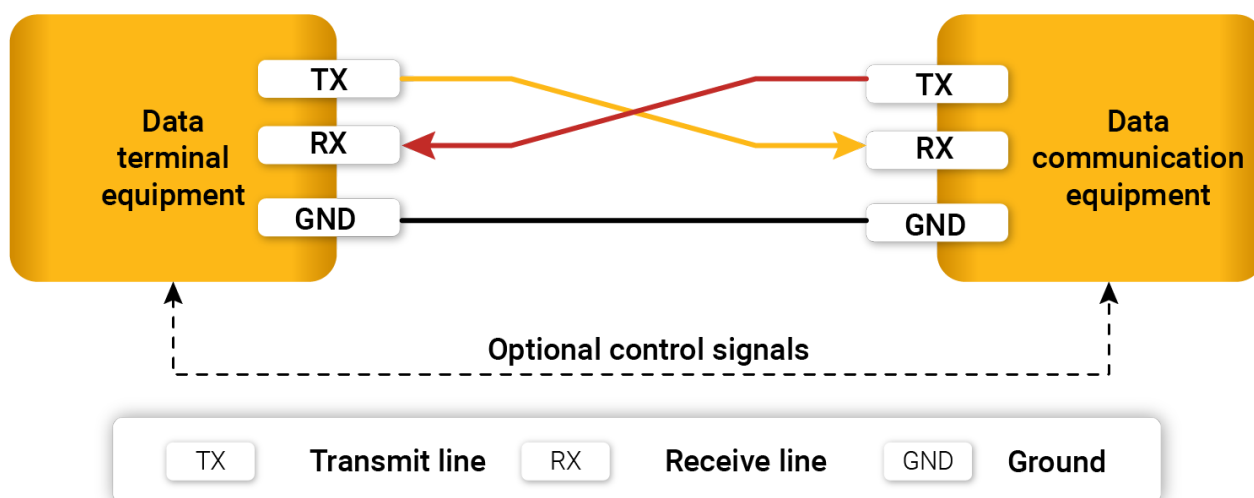
| Factors            | Details                                               |
|--------------------|-------------------------------------------------------|
| Type               | Synchronous serial audio interface                    |
| Data lines         | Serial data (SD), serial clock (CK), word select (WS) |
| Communication mode | Half-duplex (stereo audio stream)                     |
| Number of devices  | 1 master, 1 or more slaves                            |

| Factors       | Details                                                                                    |
|---------------|--------------------------------------------------------------------------------------------|
| Speed         | Common sample rates: 44.1 kHz, 48 kHz, 96 kHz, 192 kHz<br>(bit rate depends on resolution) |
| Distance      | Short; typically board-level (a few centimeters to tens of centimeters)                    |
| Applications  | Digital audio codecs, DACs, ADCs, sound processors<br>MP3 players, TVs                     |
| Advantages    | Low jitter in PCB audio signals, low pin count, widely supported in audio ICs              |
| Disadvantages | No built-in error correction, limited to audio data, not hot-pluggable                     |

## 4.5 RS-232

RS-232 is a standard for serial communication using single-ended signaling, typically for point-to-point connections over short distances.

### RS-232 serial link



RS-232 serial link for point-to-point communication between a data terminal equipment (DTE) and data communication equipment (DCE)

**Table 5:** RS-232 interface features

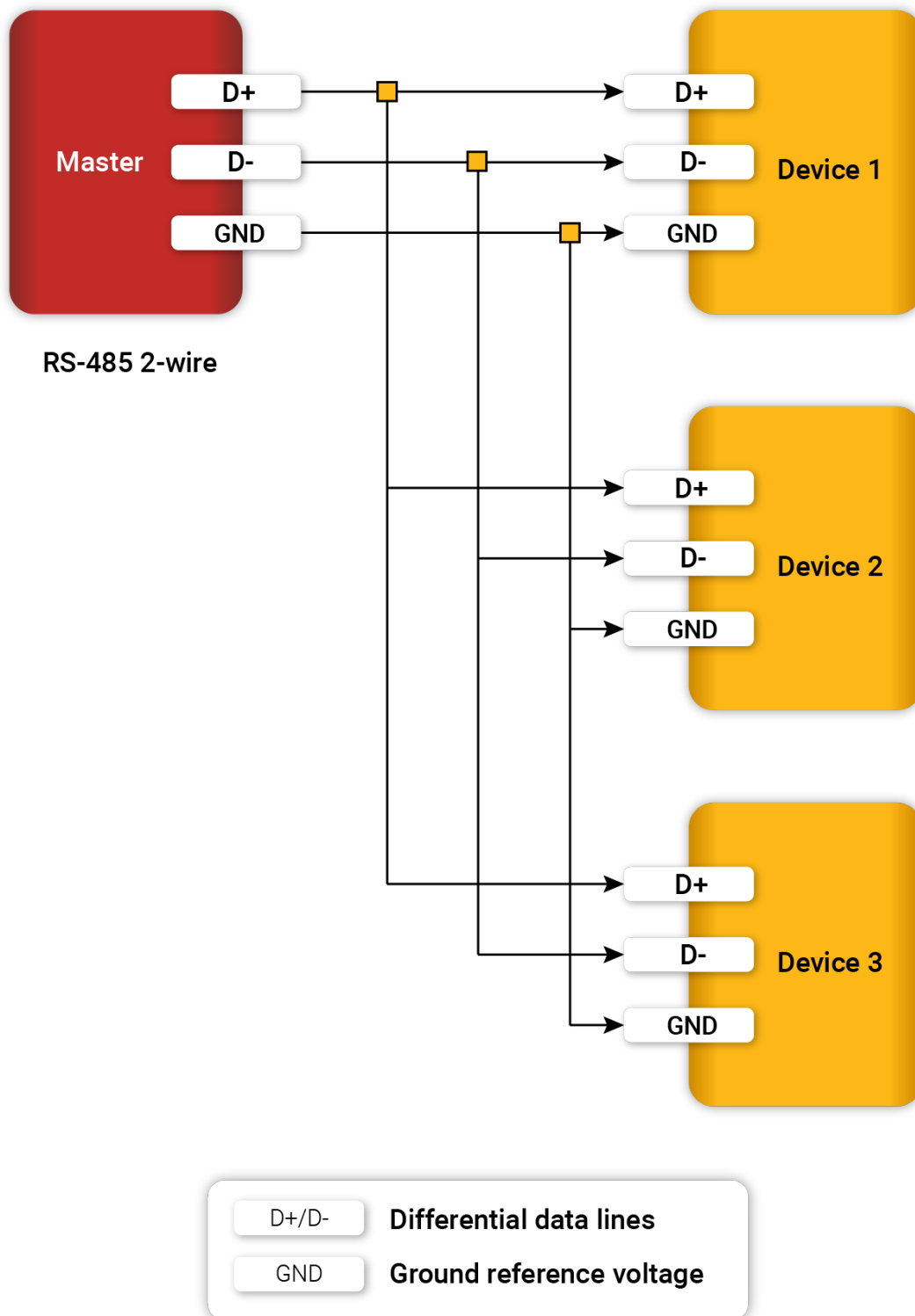
| Factors            | Details                                                                           |
|--------------------|-----------------------------------------------------------------------------------|
| Type               | Asynchronous serial communication                                                 |
| Data lines         | Transmit (TX), receive (RX), ground (GND)                                         |
| Communication mode | Full-duplex (with control signal)                                                 |
| Number of devices  | 2                                                                                 |
| Speed              | Up to 1 Mbps                                                                      |
| Distance           | Up to 15 meters at lower baud rate; shorter at higher speeds                      |
| Applications       | Industrial equipment, legacy systems, point-of-sale (POS) systems                 |
| Advantages         | Simple, widely supported, good for short-distance communication                   |
| Disadvantages      | Lower speed, short range compared to RS-485, bulk connectors, point-to-point only |

## 4.6 RS-485

It is a serial communication standard that uses differential signaling. RS-485 supports multiple devices on a bus over longer distances.



## RS-485 multi-device bus layout



RS-485 serial communication interface using a differential pair

**Table 6: RS-485 interface features**

| Factors            | Details                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------|
| Type               | Asynchronous serial communication                                                          |
| Data lines         | Differential pairs (D+ /D-)                                                                |
| Communication mode | Half-duplex (2-wire) or Full-duplex (4-wire)                                               |
| Number of devices  | Up to 32 (can be extended with special transceivers)                                       |
| Speed              | Up to 10 Mbps for short distances; lower speeds for longer distances                       |
| Distance           | Up to 1200 meters at 100 kbps (speed decreases as distance increases)                      |
| Applications       | Industrial machines, legacy systems, POS systems, and building automation                  |
| Advantages         | Robust against noise, supports long-distance communication, multi-drop capable             |
| Disadvantages      | Lower data rates compared to modern standards, and the connectors are bulky in some setups |

Use Sierra Circuits' Component Selector to choose the right electronic parts for your PCBAs.



**PCB DESIGN TOOL**

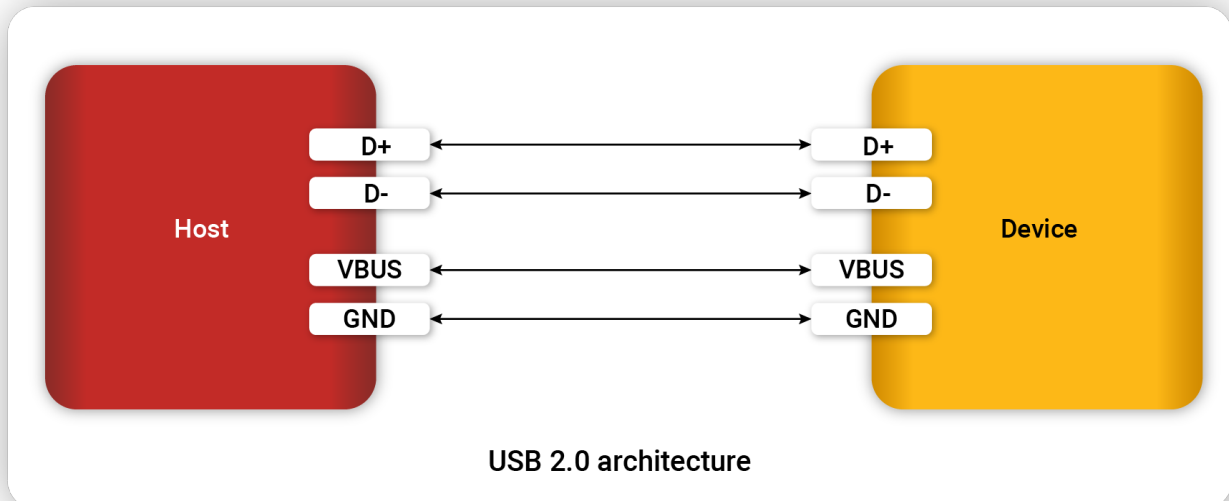
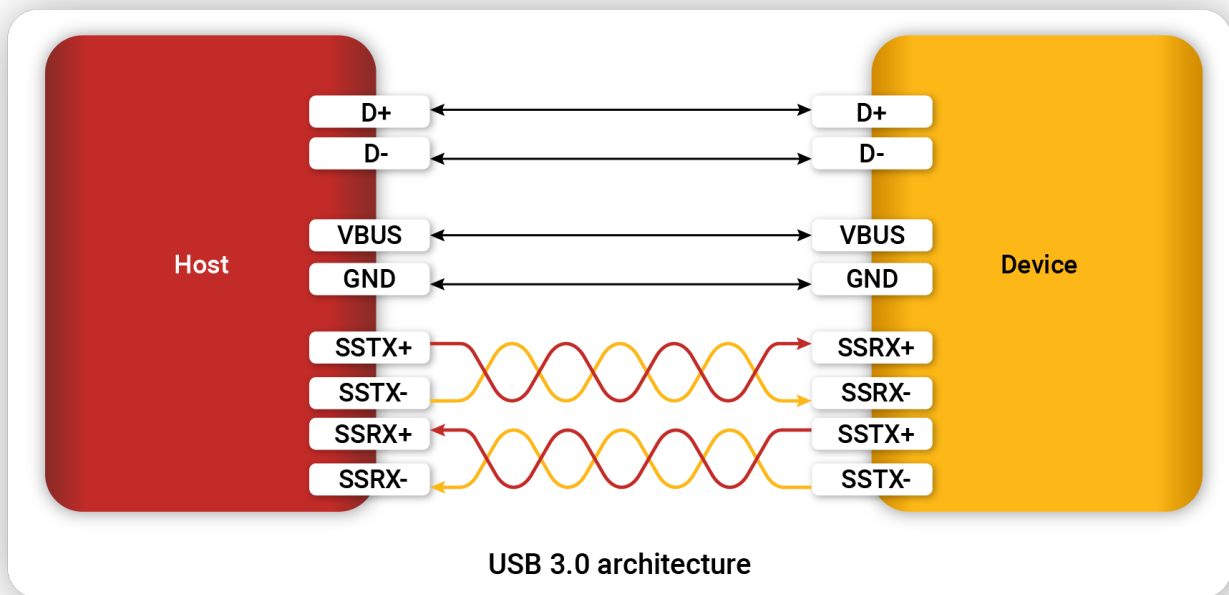
**Component Selector**

 **TRY TOOL**

**4.7 Universal serial bus (USB)**

It is a standardized wired protocol used to connect peripherals to computers, supporting plug-and-play and data/power transfer over a four-wire interface.

## USB host–device interaction



|         |                                               |
|---------|-----------------------------------------------|
| D+/D-   | Differential data lines                       |
| VBUS    | Voltage bus for power supply                  |
| GND     | Ground reference voltage                      |
| SSTX+/- | SuperSpeed transmit lines (differential pair) |
| SSRX+/- | SuperSpeed receive lines (differential pair)  |

Data flow between a host and a connected device over USB 3.0 and USB 2.0 PCB interfaces

**Table 7:** USB interface features

| Factors            | Details                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------|
| Type               | Plug-and-play serial communication                                                                    |
| Data lines         | USB 2.0: D+, D-, VBUS, GND<br>USB 3.0: D+, D-, VBUS, GND, SSRX+, SSRX-, SSTX+, SSTX-                  |
| Communication mode | USB 2.0: Half-duplex (host-controlled, polling)<br>USB 3.0: Dual-simplex                              |
| Number of devices  | Up to 127 devices per host                                                                            |
| Speed              | Up to 40 Gbps (USB4), 5 Gbps+ (USB 3.0),<br>480 Mbps (USB 2.0), 12 Mbps (USB 1.1), 1.5 Mbps (USB 1.0) |
| Distance           | Up to 5 meters typically                                                                              |
| Applications       | Computers, smartphones, peripherals, storage devices                                                  |
| Advantages         | Widely supported, hot-pluggable, high-speed                                                           |
| Disadvantages      | Cable length limitations                                                                              |

Use Sierra Circuits' Bandwidth, Rise Time and Critical Length Calculator to compute 3dB BW, signal rise time, and critical length of your signal.



**PCB DESIGN TOOL**

**Bandwidth, Rise Time and Critical Length Calculator**

 [TRY TOOL](#)

4.8 Ethernet

Ethernet is a widely used networking protocol for wired local area networks (LANs), using twisted-pair cables to transmit data in frames based on the IEEE 802.3 standard.

## Ethernet network interface



Ethernet communication interfaces highlighting key variants—fast, gigabit, and single pair connections

**Table 8:** Ethernet interface features

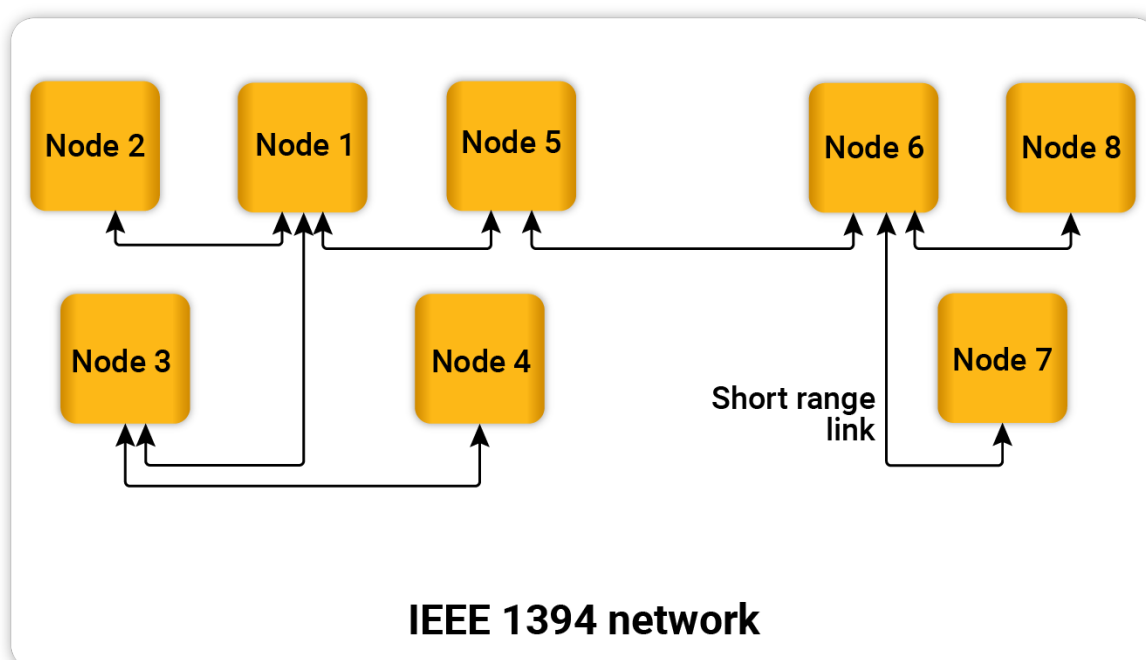
| Factors            | Details                                                |
|--------------------|--------------------------------------------------------|
| Type               | Wired network communication                            |
| Data lines         | Twisted pair (e.g., CAT5, CAT6), or fiber optic cables |
| Communication mode | Full-duplex                                            |
| Number of devices  | Scalable (depends on switching infrastructure)         |

| Factors       | Details                                                |
|---------------|--------------------------------------------------------|
| Speed         | Wired network communication                            |
| Distance      | Twisted pair (e.g., CAT5, CAT6), or fiber optic cables |
| Applications  | LANs, IoT devices, servers, and industrial automation  |
| Advantages    | Reliable, high-speed, standardized                     |
| Disadvantages | Requires infrastructure (cabling, routers, switches)   |

## 4.9 FireWire (IEEE 1394)

FireWire is a high-speed, serial communication protocol developed by Apple and standardized as IEEE 1394. It supports isochronous and asynchronous data transfer and has been widely used in multimedia applications.

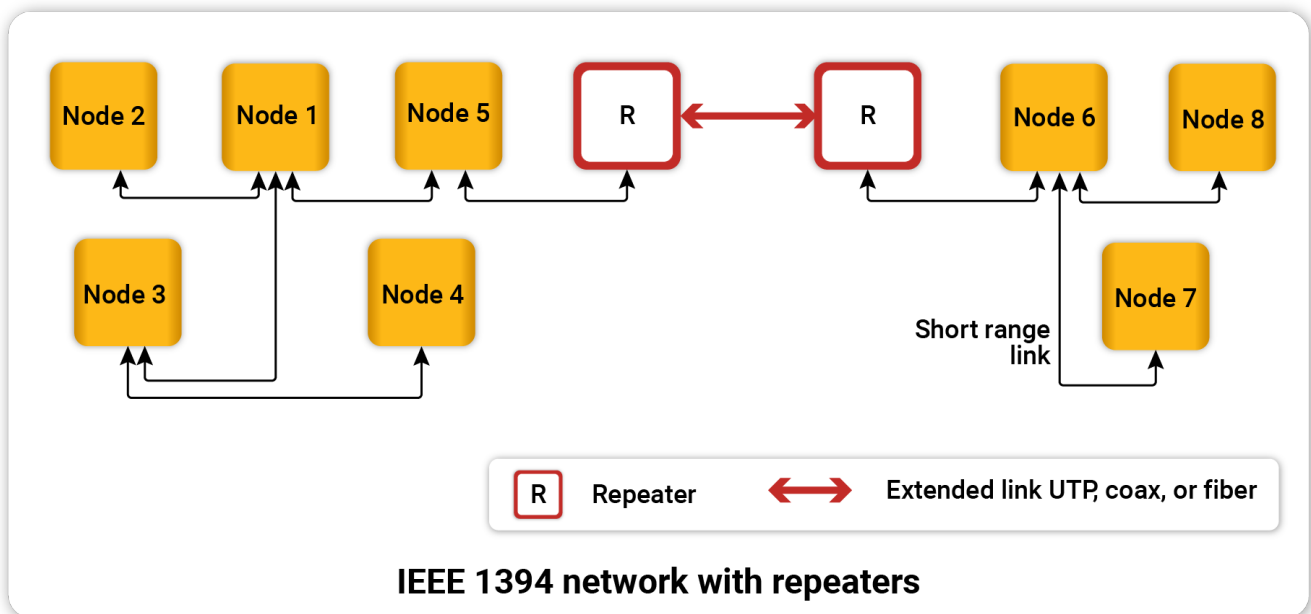
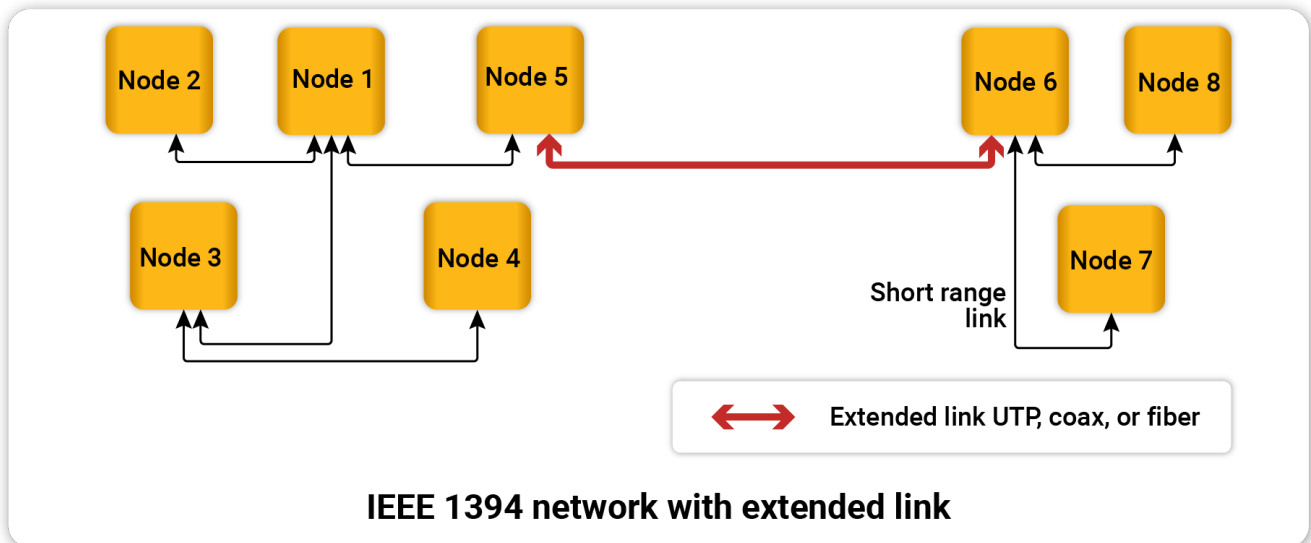
### FireWire peer-to-peer chain



Peer-to-peer serial communication topology enabled by the FireWire interface

FireWire allows peer-to-peer communication, enabling devices to talk to each other without a host (unlike USB). It also supports hot-swapping and real-time data transfer.

### FireWire peer-to-peer network with extended link



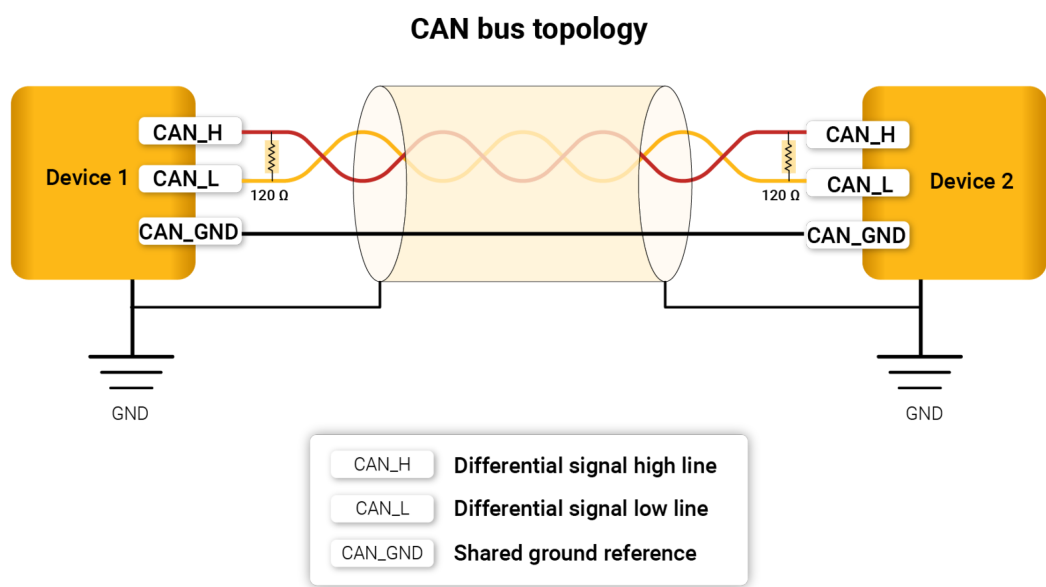
The short-distance of FireWire connection is extended using unshielded twisted pair (UTP), coaxial cable, or optical fiber to accommodate longer reach between devices

**Table 9:** FireWire interface features

| Factors            | Details                                                                             |
|--------------------|-------------------------------------------------------------------------------------|
| Type               | High-speed serial bus                                                               |
| Data lines         | Twisted-pair differential pairs for data and strobe; control signals                |
| Communication mode | Full-duplex                                                                         |
| Number of devices  | Up to 63 devices on a bus                                                           |
| Speed              | IEEE 1394a: 100, 200, 400 Mbps; IEEE 1394b: 800, 1600, 3200 Mbps                    |
| Distance           | Up to 4.5 meters per cable (IEEE 1394a); up to 100 meters with optical (IEEE 1394b) |
| Applications       | Digital video (DV), audio interfaces, external hard drives, industrial systems      |
| Advantages         | Peer-to-peer architecture, high throughput, real-time isochronous support           |
| Disadvantages      | Costlier than USB, gradually phased out in favor of USB and Thunderbolt             |

4.10 Controller area network (CAN)

The [CAN communication protocol](#) is a multi-master serial system commonly used in automotive and industrial applications for real-time data exchange over a two-wire bus.



CAN communication over a single twisted pair with 120 Ω termination at both ends



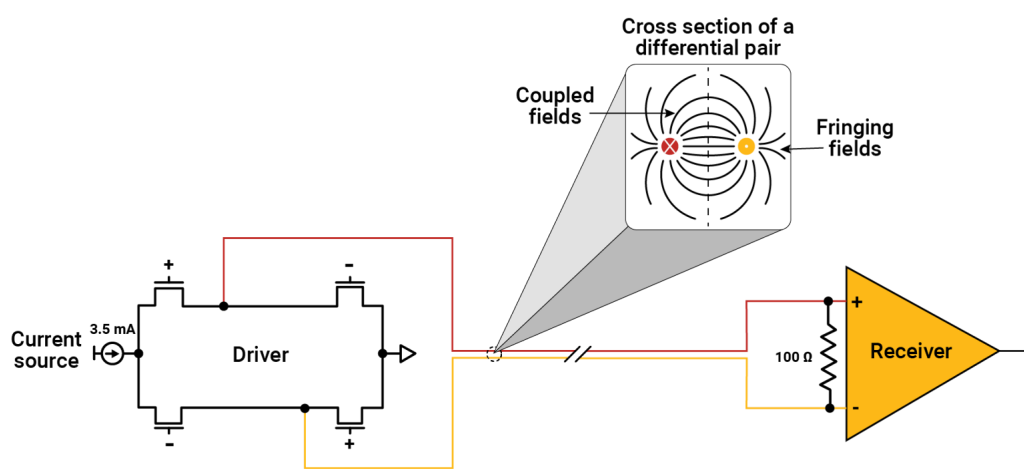
**Table 10:** CAN interface features

| Factors            | Details                                                                            |
|--------------------|------------------------------------------------------------------------------------|
| Type               | Multi-master serial communication                                                  |
| Data lines         | CAN_H, CAN_L (twisted-pair cable)                                                  |
| Communication mode | Half-duplex (broadcast-based)                                                      |
| Number of devices  | Typically up to 112 devices                                                        |
| Speed              | Up to 1 Mbps (CAN), 5 Mbps (CAN-FD)                                                |
| Distance           | Up to 40 meters at max speed; longer at lower speeds                               |
| Applications       | Automotive, robotics, industrial automation                                        |
| Advantages         | Robust error handling, reliable in noisy environments, and real-time communication |
| Disadvantages      | Limited data rate compared to Ethernet                                             |

**4.11 Low-voltage differential signaling (LVDS)**

It is a high-speed, low-power signaling method that uses differential pairs to transmit data with low noise and electromagnetic interference, making it suitable for [routing high-speed PCBs](#).

**LVDS signal transmission**



**Point-to-point high-speed communication between the transmitter and the receiver using an LVDS interface**

**Table 11: LVDS interface features**

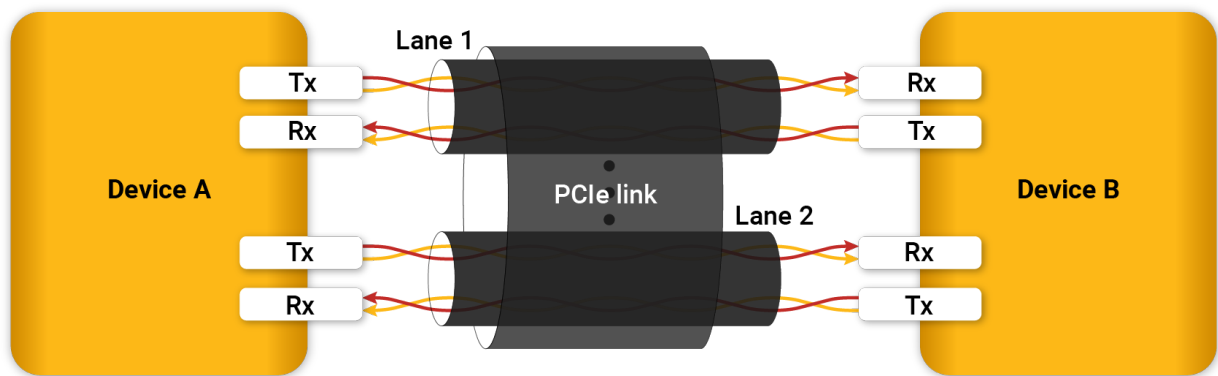
| Factors            | Details                                               |
|--------------------|-------------------------------------------------------|
| Type               | High-speed differential signaling                     |
| Data lines         | Differential pairs                                    |
| Communication mode | Full-duplex or simplex (depending on design)          |
| Number of devices  | 2 (point-to-point)                                    |
| Speed              | Hundreds of Mbps                                      |
| Distance           | Several meters (depends on quality)                   |
| Applications       | Displays, high-speed cameras, medical devices, memory |
| Advantages         | Low noise, low power, high speed                      |
| Disadvantages      | Requires careful impedance matching, higher cost      |

Try Sierra Circuits' Impedance Calculator to precisely model your high-speed traces and prevent signal distortion in your designs.

## 4.12 Peripheral component interconnect express (PCIe)

PCIe is a high-speed serial expansion bus standard that connects internal components like GPUs and SSDs to a motherboard.

PCIe lane structure



Communication between devices using a PCIe interface over differential pairs

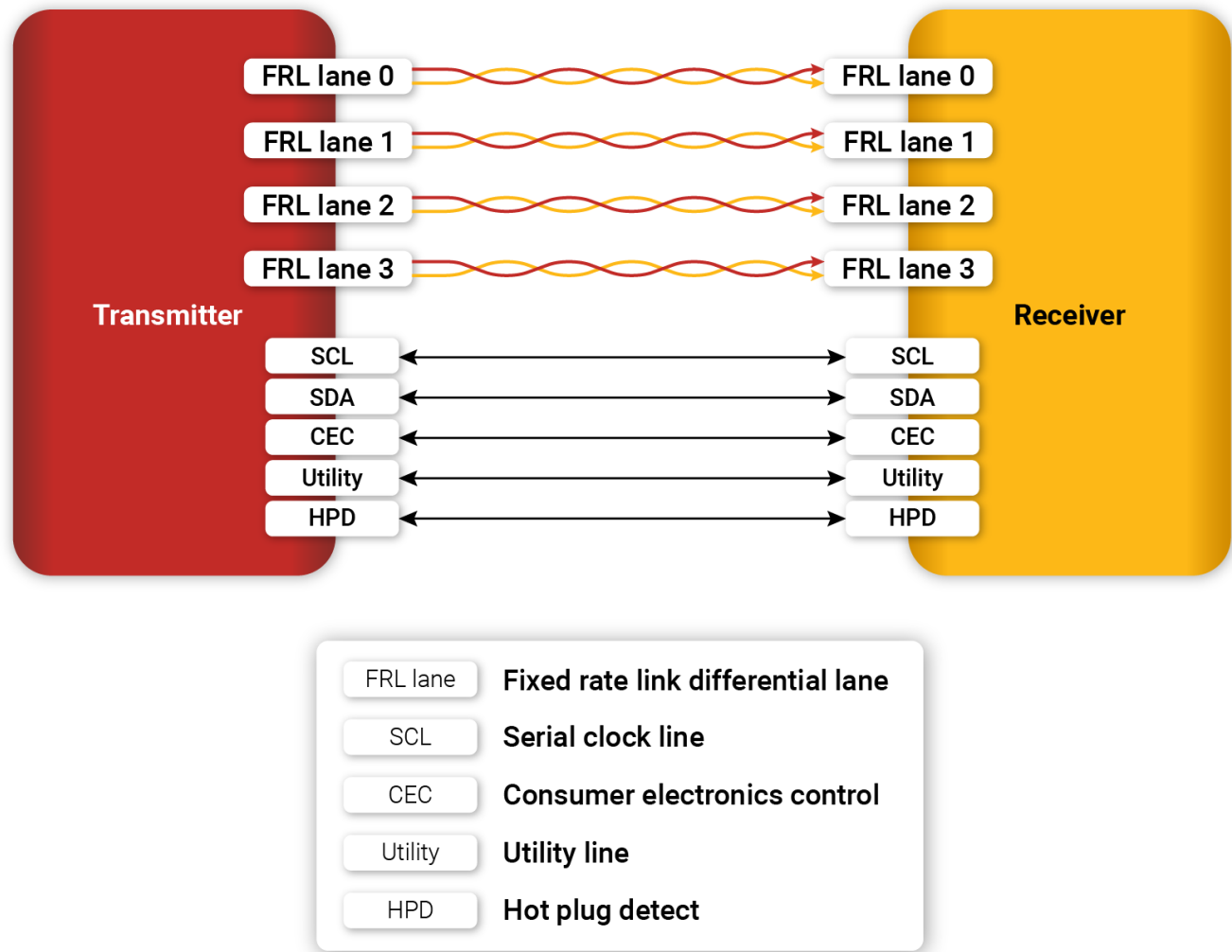
Table 12: PCIe interface features

| Factors            | Details                                       |
|--------------------|-----------------------------------------------|
| Type               | High-speed serial expansion bus               |
| Data lines         | Differential pairs (lanes)                    |
| Communication mode | Full-duplex (lane pairs)                      |
| Number of devices  | Typically, 1 root and multiple endpoints      |
| Speed              | Up to 64 GT/s (PCIe 6.0), 16 GT/s (PCIe 4.0)  |
| Distance           | Short (inside PCBs)                           |
| Applications       | Graphics cards, SSDs, and network cards       |
| Advantages         | High throughput, scalable lanes (x1, x4, x16) |
| Disadvantages      | PCB complexity, short range only              |

### 4.13 High-definition multimedia interface (HDMI)

This is a popular digital interface for transmitting uncompressed video and audio data between devices like monitors and media players over a single cable.

#### HDMI fixed rate link (FRL) communication



HDMI communication shows unidirectional transmission of high-speed digital signals from the transmitter to the receiver

Table 13: HDMI features

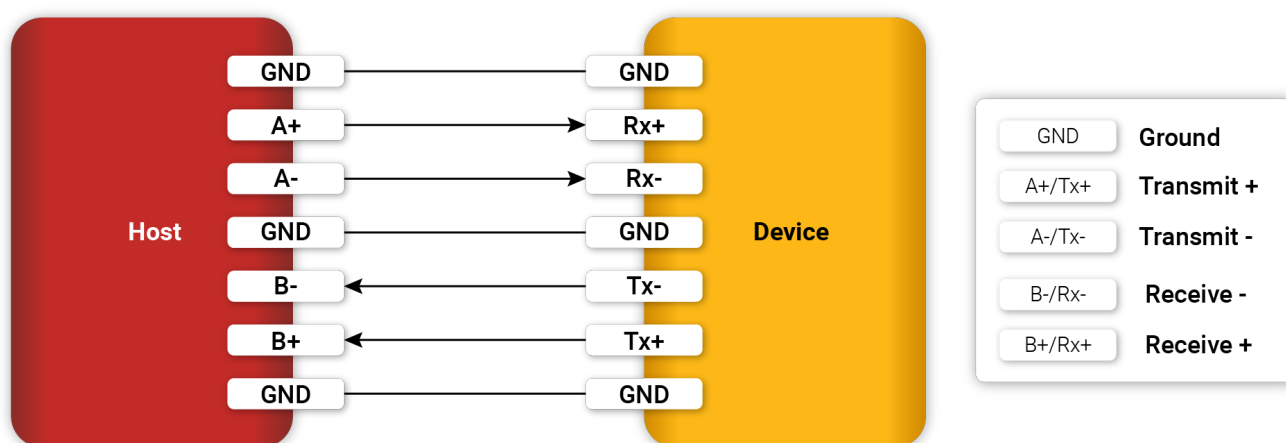
| Factors            | Details                                                      |
|--------------------|--------------------------------------------------------------|
| Type               | Digital multimedia interface                                 |
| Data lines         | Multiple differential pairs (3 or 4 for data), control lines |
| Communication mode | Simplex (one-way: source → display)                          |

| Factors           | Details                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------|
| Number of devices | 1 source to 1 sink (point-to-point)                                                             |
| Speed             | Bandwidth: up to 48 Gbps (FRL)<br>Effective data rate: up to 42.6 Gbps (after 16b/18b encoding) |
| Distance          | 5 to 10 meters at high speeds, up to 15 meters with signal boosters                             |
| Applications      | Televisions, HDMI monitors, projectors, and gaming consoles                                     |
| Advantages        | Single cable for video, audio, and control signals                                              |
| Disadvantages     | Distance and bandwidth limitations at high resolutions, unidirectional                          |

## 4.14 Serial advanced technology attachment (SATA)

It is a computer bus interface for connecting mass storage devices like hard drives and SSDs using serial signaling for fast data transfer.

**SATA host-device link**



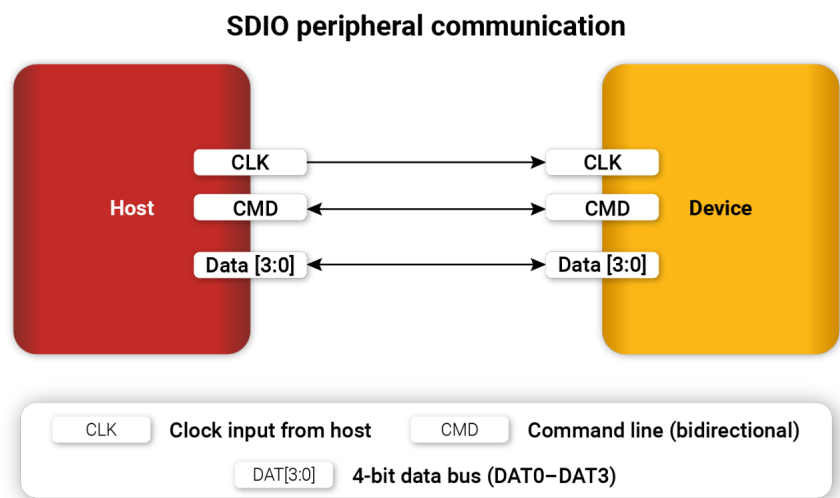
SATA communication between the host and device via differential signal pairs

**Table 14:** SATA interface features

| Factors            | Details                                             |
|--------------------|-----------------------------------------------------|
| Type               | High-speed data transfer for storage devices        |
| Data lines         | Differential pair                                   |
| Communication mode | Full-duplex                                         |
| Number of devices  | 1-to-1 per port                                     |
| Speed              | Up to 6 Gbps (SATA III)                             |
| Distance           | Up to 1 meter (typically)                           |
| Applications       | Hard drives, SSDs                                   |
| Advantages         | High-speed data transfer for storage, hot-swappable |
| Disadvantages      | Short range, power consumption                      |

4.15 Secure digital input output (SDIO)

This wired interface type is an extension of the SD card standard that allows peripheral devices to interface with a host via the SD card slot using serial communication.

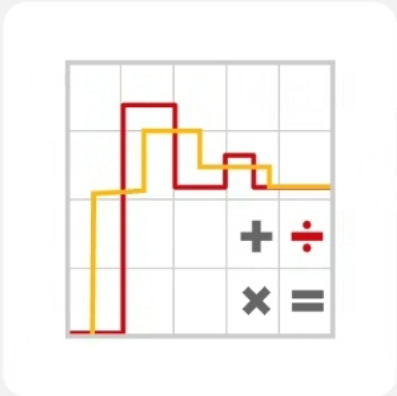


Communication between the host and device through an SDIO interface for synchronous serial data exchange

**Table 15:** SDIO interface features

| Factors            | Details                                                                             |
|--------------------|-------------------------------------------------------------------------------------|
| Type               | Synchronous serial peripheral connection based on the SD card standard              |
| Data lines         | CLK, CMD, DAT[3:0]                                                                  |
| Communication mode | Half-duplex                                                                         |
| Number of devices  | Typically 1 host, 1 device (some advanced host controllers support up to 7 devices) |
| Speed              | Up to 25 MB/s, Up to 104 Mbps (SDIO 3.0)                                            |
| Distance           | Short (within device or peripheral)                                                 |
| Applications       | Wi-Fi modules, GPS, Bluetooth peripherals                                           |
| Advantages         | Compact, low power                                                                  |
| Disadvantages      | Lower speeds than PCIe/USB                                                          |

Sierra Circuits' Transmission Line Reflection Calculator helps you understand the signal characteristics of your PCB during the design phase.



**PCB DESIGN TOOL**

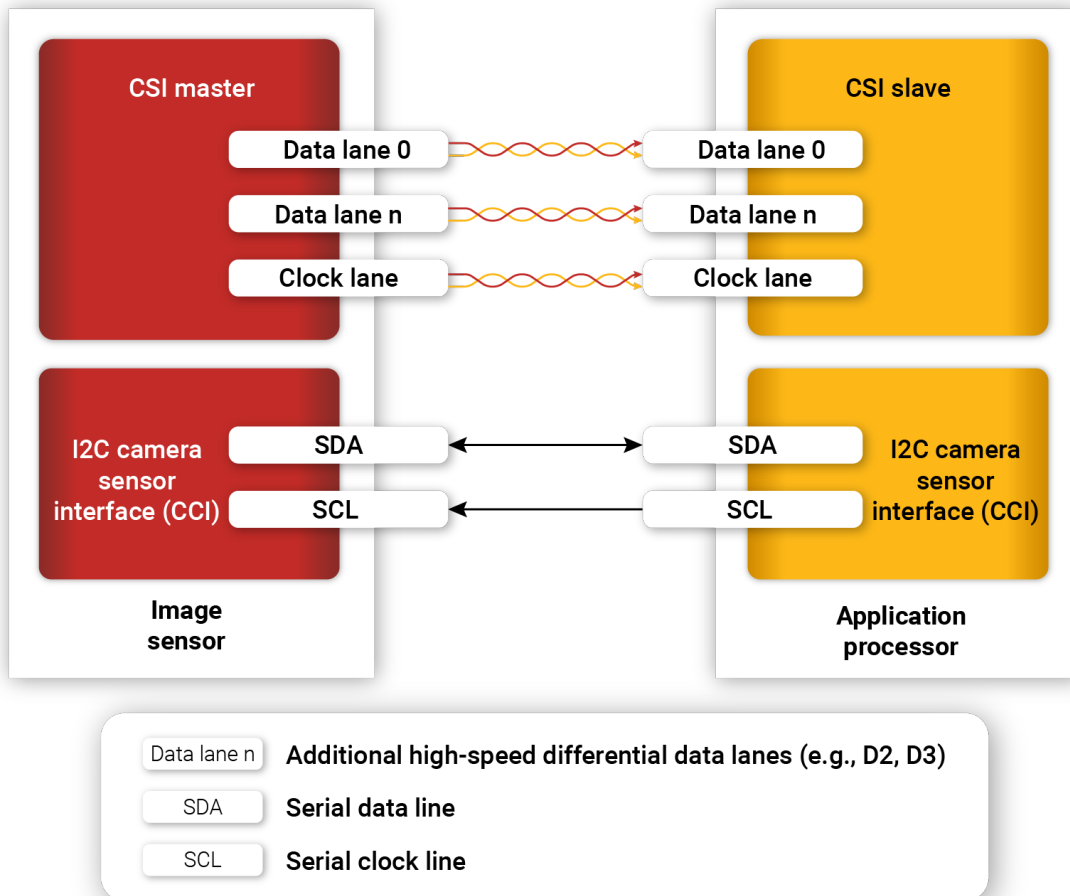
**Transmission Line Reflection Calculator**

 **TRY TOOL**

4.16 Mobile industry processor interface (MIPI)

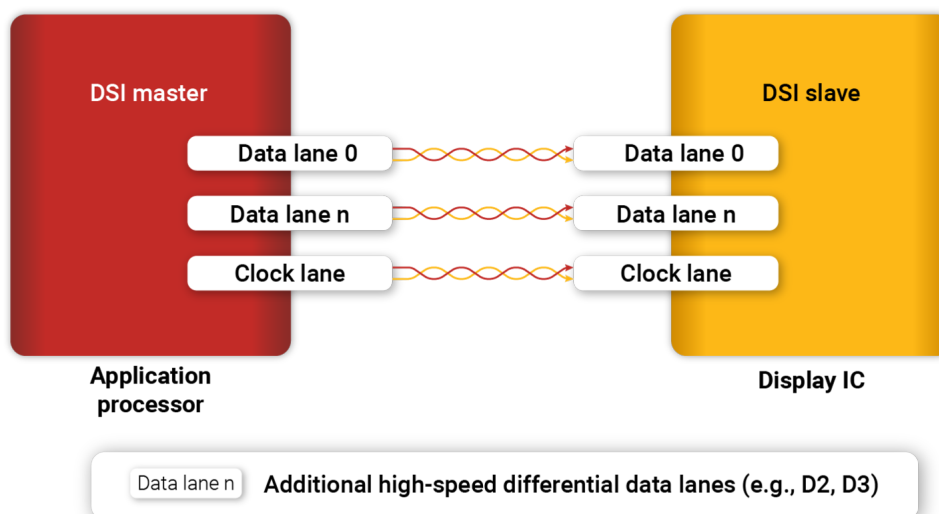
MIPI defines low-power, high-speed serial interfaces, such as camera serial interface (CSI) and display serial interface (DSI). It is used for connecting cameras and displays to processors in mobile and embedded systems using differential data lanes.

## MIPI CSI-2 connection



MIPI CSI communication between an image sensor and a processor via differential high-speed data and clock lanes

## MIPI DSI connection



The MIPI DSI is used for communication between the application processor and display panel



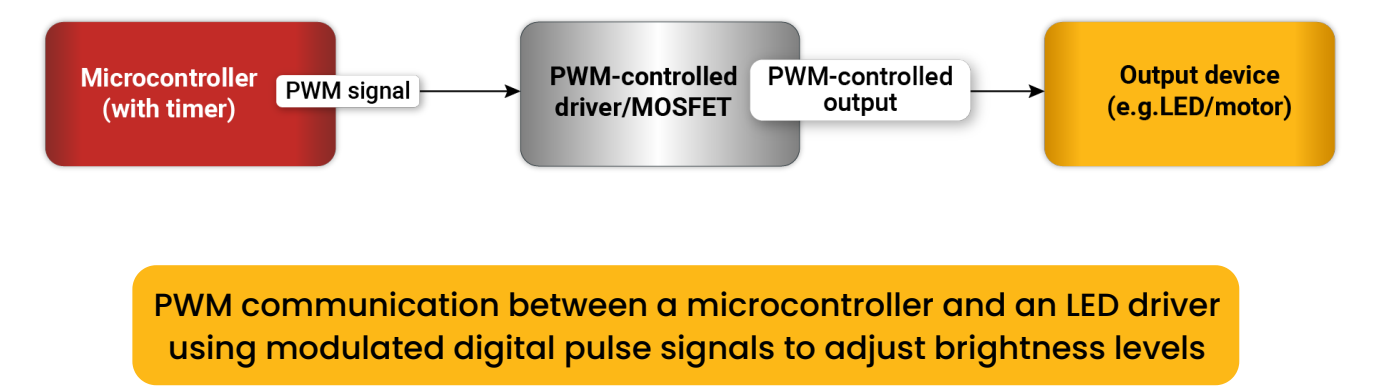
**Table 16:** MIPI features

| Factors            | Details                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------|
| Type               | High-speed interface standards suite (e.g., MIPI CSI, DSI)                                               |
| Data lines         | Differential pairs (2–4 lanes per direction, via D-PHY or C-PHY)                                         |
| Communication mode | Simplex (CSI: sensor → processor, DSI: processor → display) or half-duplex (depending on implementation) |
| Number of devices  | Point-to-point (1 source to 1 sink)                                                                      |
| Speed              | Up to 6 Gbps (DSI 2.0), 3 Gbps (CSI 2.0)                                                                 |
| Distance           | Short (on-PCB or module-to-SoC)                                                                          |
| Applications       | Smartphones, tablets, embedded cameras, LCD panels                                                       |
| Advantages         | Standardized for mobile, high-speed, scalable, low-power                                                 |
| Disadvantages      | Complex design, limited range, sensitive to noise                                                        |

**4.17 Pulse width modulation (PWM)**

This is a method of transmitting analog signal levels by varying the duty cycle of a digital signal, commonly used for motor control and dimming LEDs.

**Pulse width modulation interface**



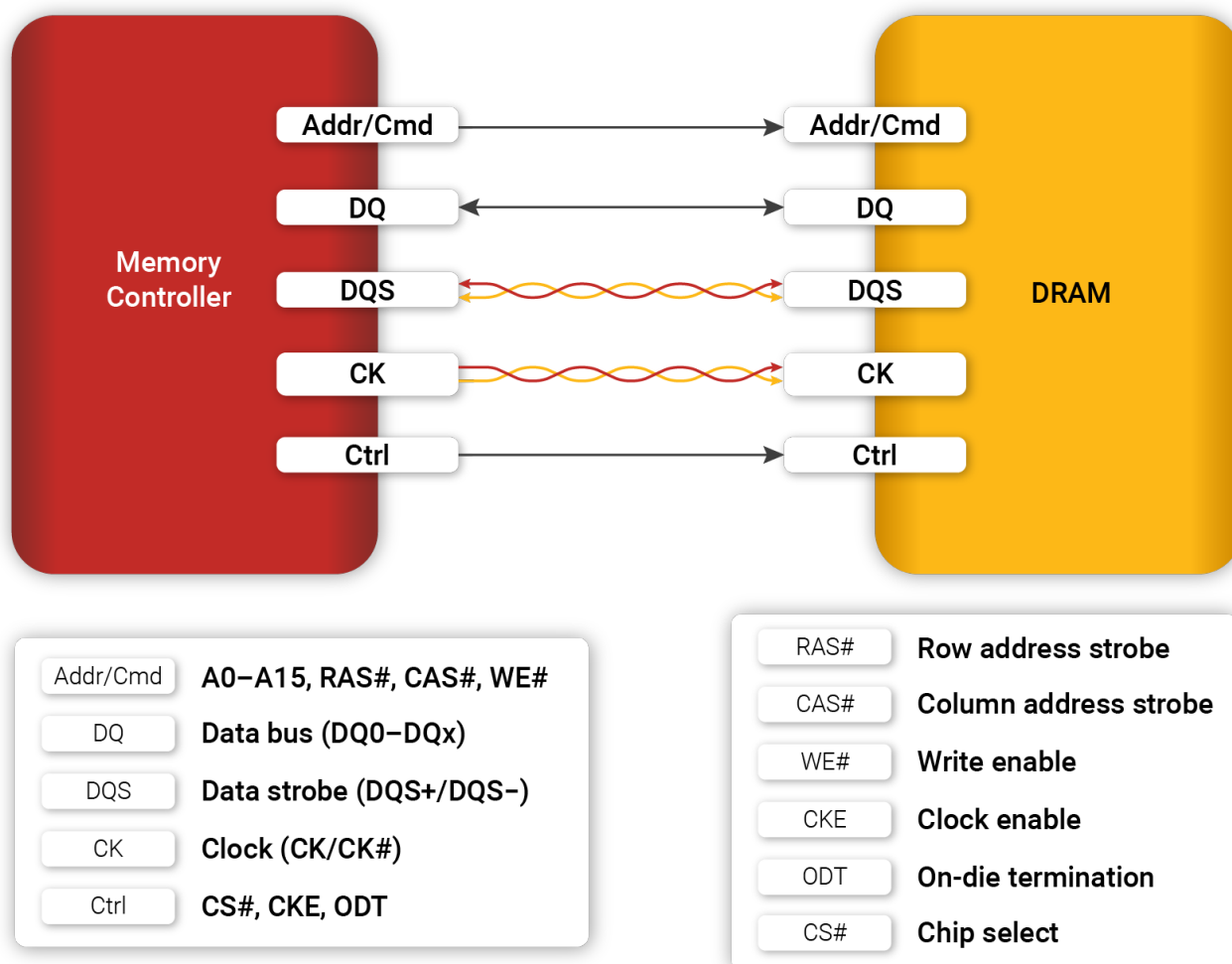
**Table 17:** PWM interface features

| Factors            | Details                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| Type               | Digital pulses                                                                                                       |
| Data lines         | Single wire (for digital control)                                                                                    |
| Communication mode | Simplex (controller to device)                                                                                       |
| Number of devices  | 1 controller to many devices (if shared)                                                                             |
| Speed              | Varies (frequency can range from 1 Hz to several MHz)                                                                |
| Distance           | Short (dependent on the device control)                                                                              |
| Applications       | Motor speed control, LED dimming, audio signal modulation                                                            |
| Advantages         | Energy-efficient, simple design                                                                                      |
| Disadvantages      | Switching noise and electromagnetic interference if not properly filtered. Non-linear behavior in some applications. |

**4.18 Double data rate (DDR)**

**DDR memory** is a high-speed synchronous interface used to connect dynamic random-access memory (DRAM) to a memory controller, commonly found in processors, system on a chip (SoC), and field-programmable gate arrays (FPGAs). It transfers data on both the rising and falling edges of the clock signal, effectively doubling the data rate compared to single data rate (SDR) memory.

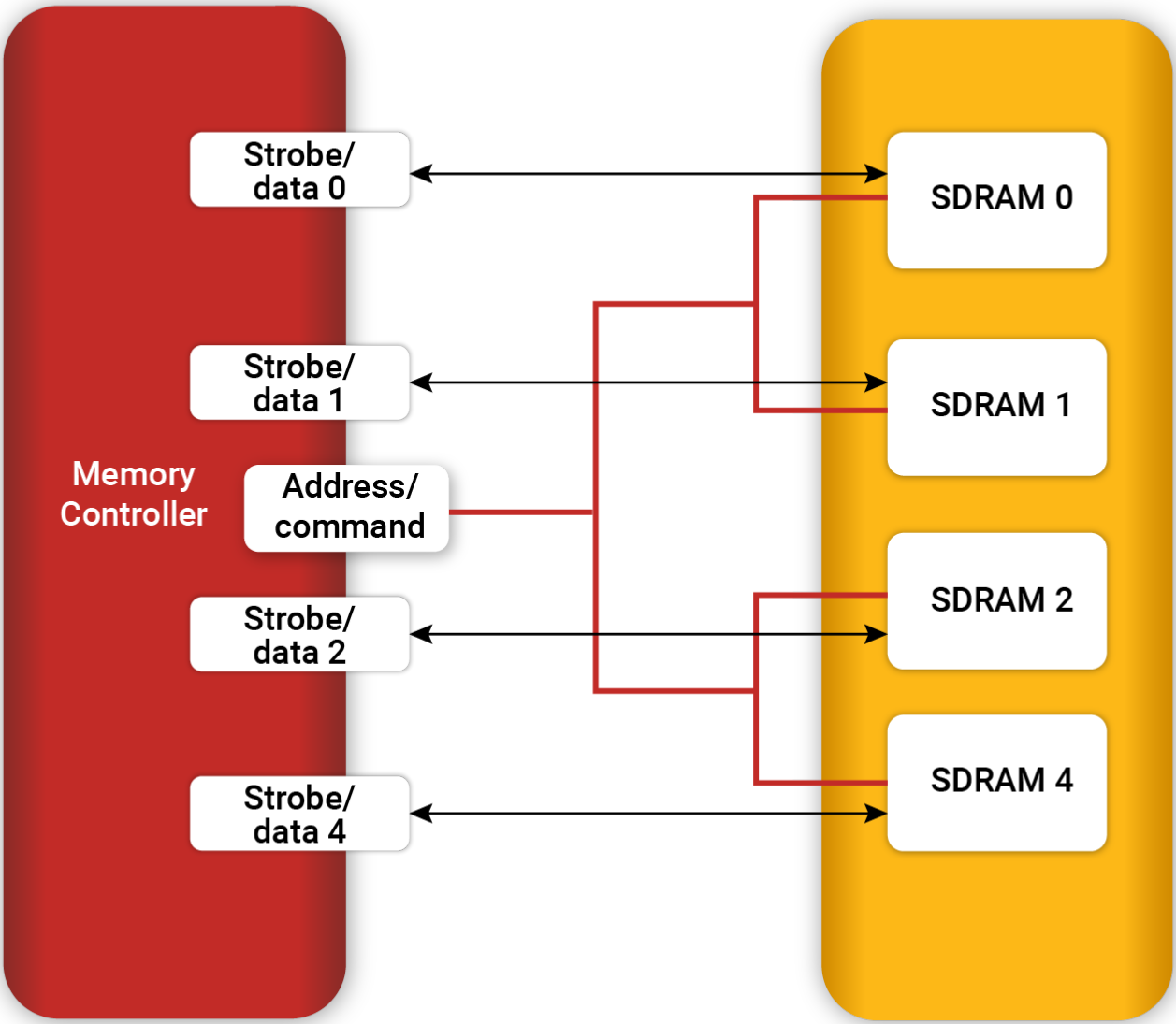
## DDR memory interface between the controller and the DRAM



Communication between the memory controller and DRAM via the DDR interface

DDR operates with strict timing and signal integrity requirements, and its interface includes multiple signal types for command, address, control, and high-speed data transfer.

# Controller-to-multiple SDRAM devices via DDR interface



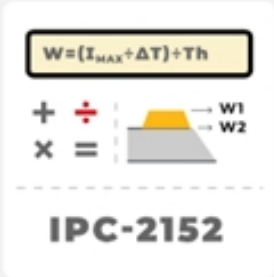
Communication between the controller and multiple SDRAM devices using the DDR PCB interface

**Table 18:** DDR interface features

| Factors            | Details                                                                 |
|--------------------|-------------------------------------------------------------------------|
| Type               | Synchronous parallel memory interface                                   |
| Data lines         | DQ (data lines), DQS (data strobe), address, command, and control lines |
| Communication mode | Half-duplex (separate read and write operations, not simultaneous)      |
| Number of devices  | 1 or more DRAM devices per channel (with or without ranks)              |

| Factors       | Details                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------|
| Speed         | DDR1: up to 400 MT/s, DDR2: up to 800 MT/s, DDR3: up to 2133 MT/s, DDR4: up to 3200 MT/s, DDR5: up to 8400+ MT/s |
| Clocking      | Double data rate (transfers data on both clock edges); external differential clock                               |
| Distance      | Very short – typically PCB trace length of a few inches (board-level only)                                       |
| Applications  | PCs, laptops, smartphones, FPGAs, embedded systems, graphics cards                                               |
| Advantages    | High bandwidth, widely supported, standardized                                                                   |
| Disadvantages | Complex routing, tight timing margins, require impedance control and termination                                 |

Try Sierra Circuits' Trace Width, Current Capacity, and Temperature Rise Calculator to accurately compute the line width and spacing for safe operation.



**PCB DESIGN TOOL**

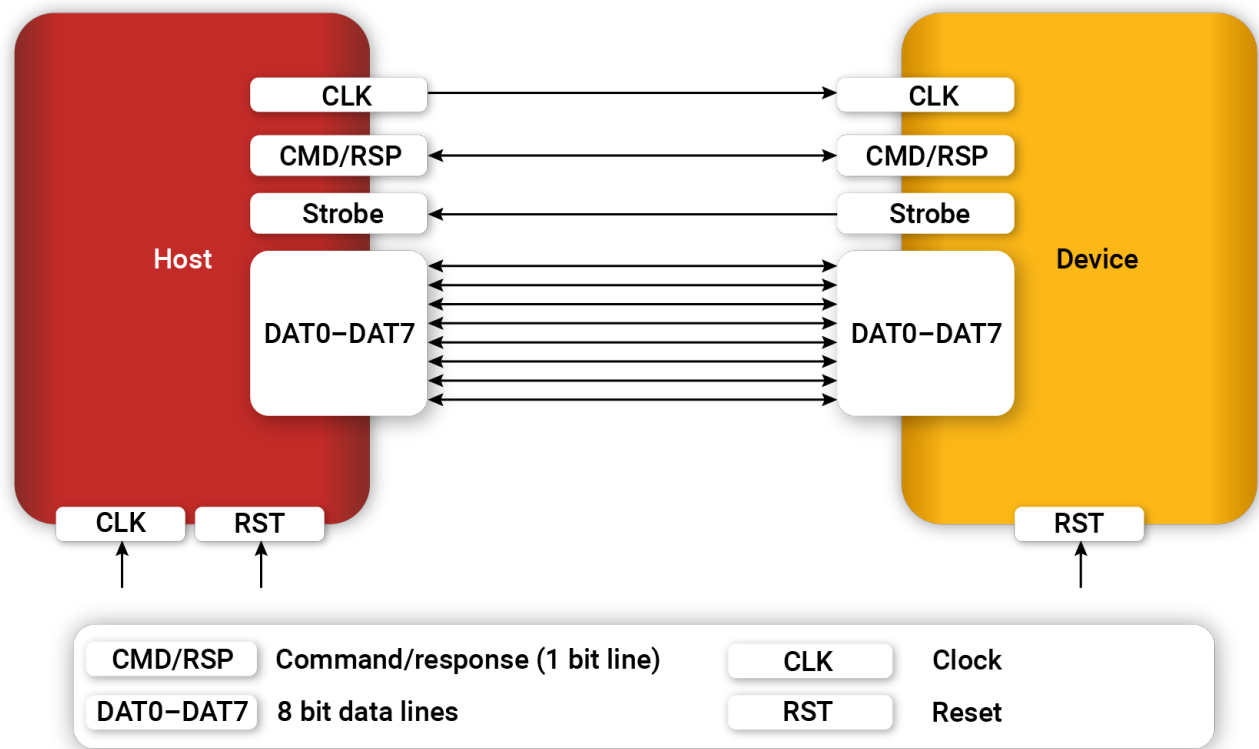
**Trace Width, Current Capacity and Temperature Rise Calculator**

 **TRY TOOL**

### 4.19 Embedded multimedia card (eMMC)

eMMC is a non-volatile storage interface based on the multimedia card (MMC) standard, commonly used in smartphones, tablets, and embedded systems. It combines a flash memory device and a controller in a single BGA package, simplifying storage integration in hardware designs.

eMMC host-device communication interface



The eMMC interface is used for communication between the host and the device via parallel signal lines

Unlike removable SD cards, eMMC is soldered directly onto the PCB and provides a managed NAND flash interface with wear leveling, bad block management, and error correction handled internally.

Table 19: eMMC interface features

| Factors            | Details                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------|
| Type               | Embedded flash memory interface (based on MMC standard)                                        |
| Data lines         | 1-bit, 4-bit, or 8-bit parallel data bus, clock, command, reset, and optionally strobe signals |
| Communication mode | Half-duplex (host initiates all transactions)                                                  |

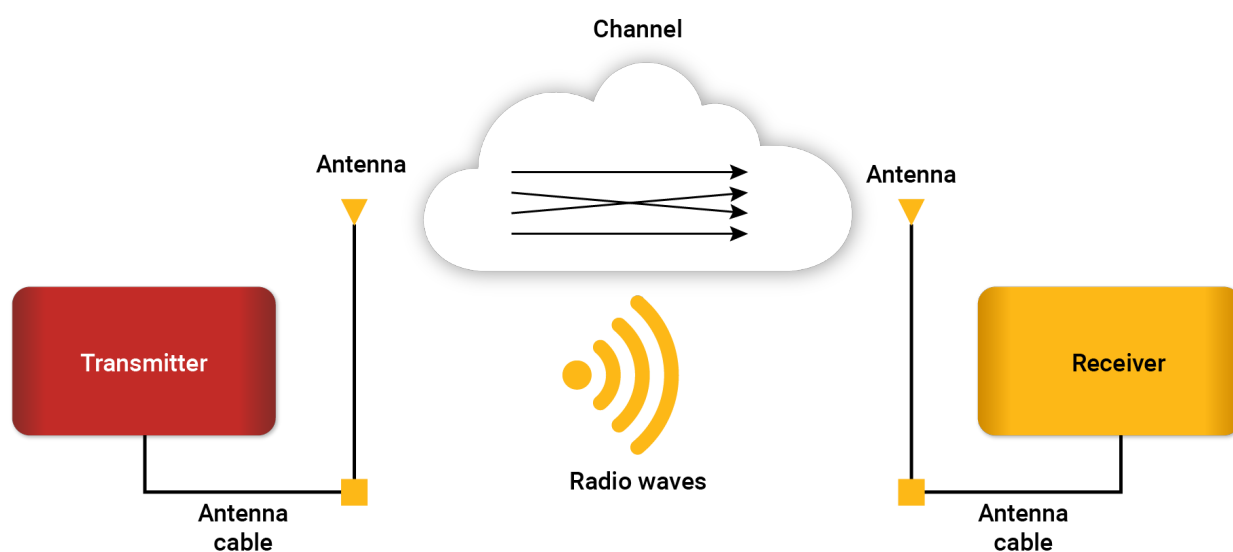
| Factors           | Details                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------|
| Number of devices | Single eMMC device per controller (not meant for sharing)                                          |
| Speed             | Up to 400 MB/s (HS400 mode)                                                                        |
| Clocking          | Synchronous clocked interface; supports dual data rate in HS200/HS400                              |
| Distance          | Very short (embedded device; traces within the PCB)                                                |
| Applications      | Smartphones, tablets, embedded systems, IoT devices, automobiles                                   |
| Advantages        | Compact, cost-effective, integrates controller and NAND flash, standard boot and storage functions |
| Disadvantages     | Not removable, limited upgrade options, lower performance than UFS                                 |

## 5. 9 types of wireless PCB interfaces

### 5.1 Wireless fidelity (Wi-Fi)

This is a wireless networking protocol based on IEEE 802.11 standards, enabling devices to connect to local area networks and the internet using radio waves.

#### Wi-Fi device connectivity



Wi-Fi communication between the PCB antenna and external devices via wireless radio frequency signals

**Table 20:** Wi-Fi interface features

| Factors            | Details                                                 |
|--------------------|---------------------------------------------------------|
| Type               | Wireless communication via radio waves                  |
| Data lines         | Radio frequency (2.4 GHz and 5 GHz, 6 GHz for Wi-Fi 6E) |
| Communication mode | Half-duplex (shared medium, collision avoidance)        |
| Number of devices  | Typically up to 250 devices per access point            |
| Speed              | Up to 9.6 Gbps (Wi-Fi 6)                                |
| Distance           | Up to 100 meters (depending on the environment)         |
| Applications       | Internet access, IoT devices, home networks             |
| Advantages         | High-speed, convenient, widely available                |
| Disadvantages      | Signal interference, security concerns                  |

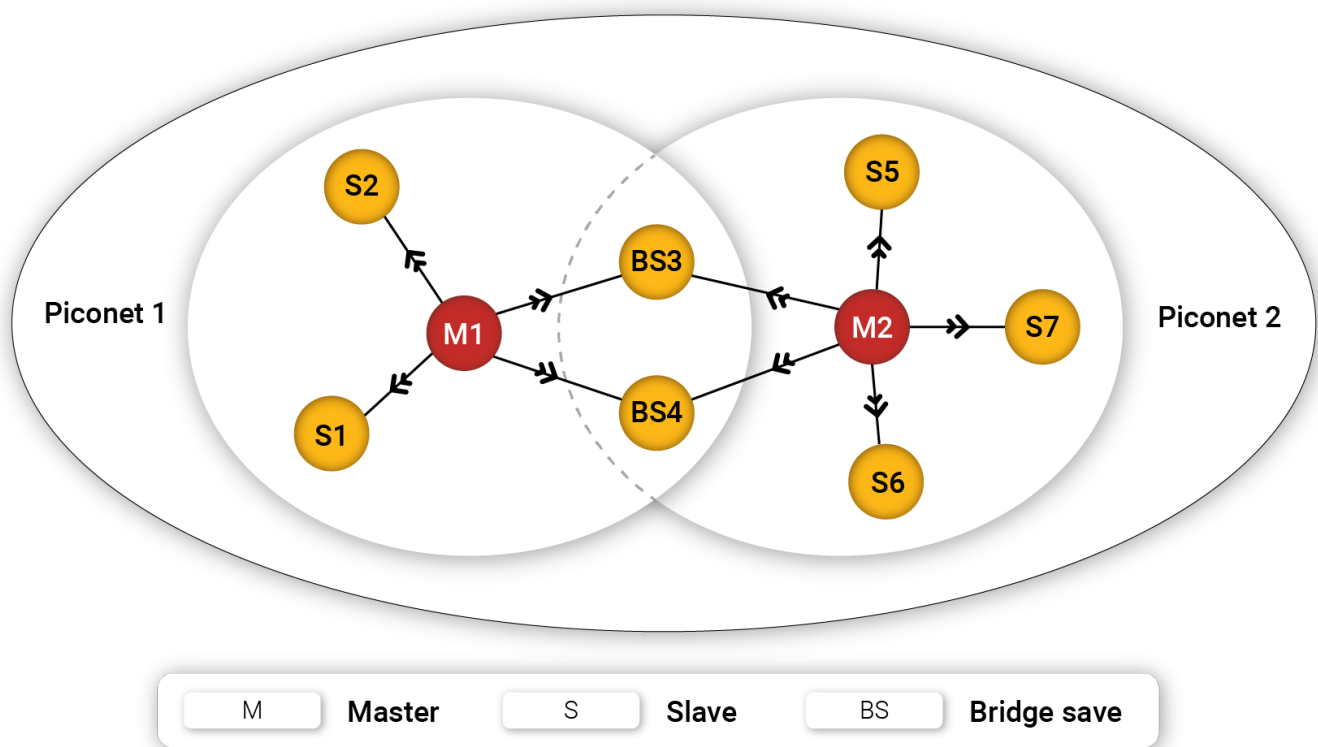
## 5.2 Bluetooth

A Bluetooth interface enables wireless communication by forming a short-range network called a piconet, which includes one master device and multiple slave devices that communicate directly with the master.

Several interconnected piconets can form a larger structure known as a scatternet, where certain devices, called bridge nodes, participate in more than one piconet to facilitate communication between them.



Bluetooth network-scatternet with two piconets



Bluetooth communication in a scatternet topology with two interconnected piconets enabling wireless master-slave device interactions

Table 21: Bluetooth interface features

| Factors            | Details                            |
|--------------------|------------------------------------|
| Type               | Short-range wireless communication |
| Data lines         | Radio frequency (2.4 GHz ISM band) |
| Communication mode | Half-duplex (TDM-based)            |
| Number of devices  | 1 master, up to 7 active slaves    |

| Factors       | Details                                                                                  |
|---------------|------------------------------------------------------------------------------------------|
| Speed         | Up to 3 Mbps (Bluetooth 2.0 + EDR), 1 Mbps (Bluetooth 4.0 LE), 2 Mbps (Bluetooth 5.0 LE) |
| Distance      | Typically up to 10 meters (classic Bluetooth), up to 100 meters (Bluetooth 5.0 LE)       |
| Applications  | Wireless audio, IoT devices, mobile peripherals                                          |
| Advantages    | Low power consumption, easy pairing                                                      |
| Disadvantages | Limited range, interference with other devices                                           |

Sierra Circuits manufactures high-precision RF and microwave PCBs with tightly controlled impedance and low signal loss. See RF and microwave PCB capabilities to learn more.

### 5.3 ZigBee (802.15.4)

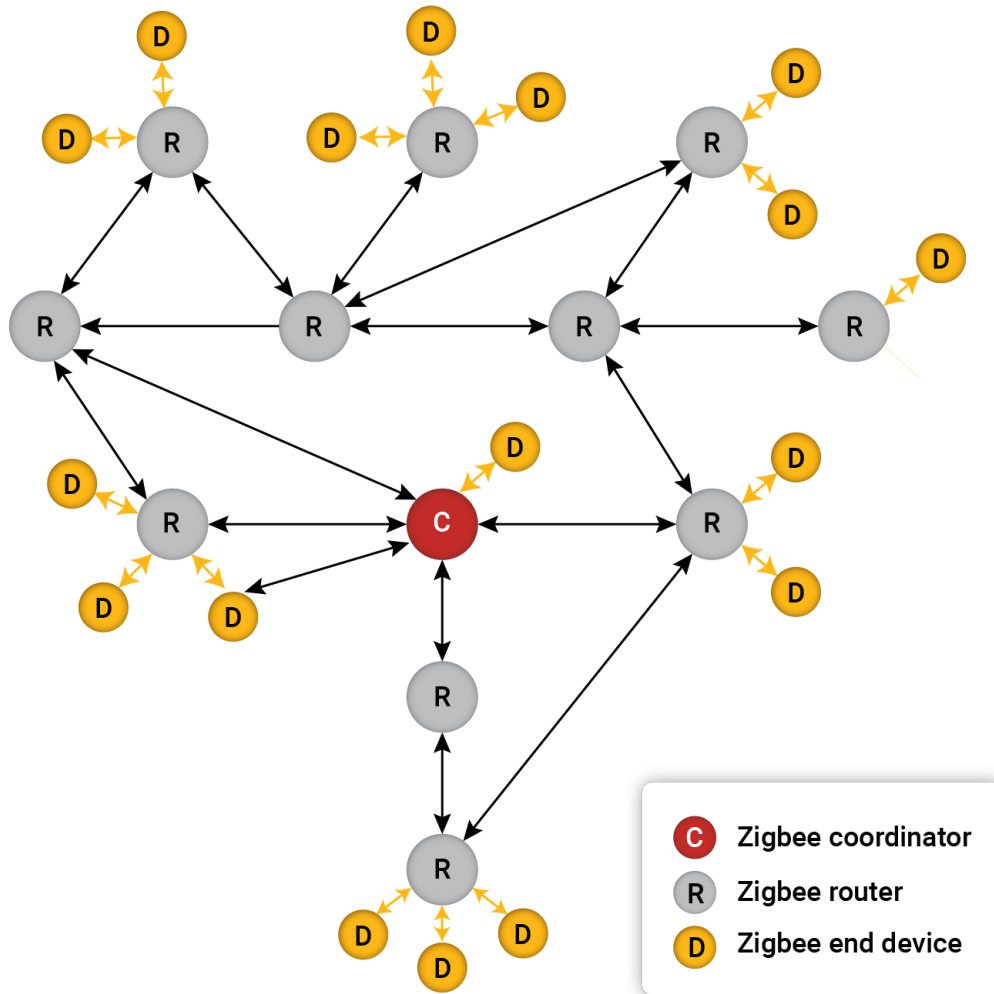
In ZigBee communication, devices form a mesh network where data is transmitted between nodes using short-range, low-power wireless signals.

The network includes the following devices and nodes.

- Coordinator- Central device that initiates, manages, and secures the network. Only one exists per network.
- Routers (nodes) - Forward data between devices, extending network range.
- End devices- Send and receive data via a parent router.

Messages can hop through multiple nodes to reach their destination, enabling reliable communication even over longer distances or around obstacles.

## ZigBee (802.15.4) mesh network



Wireless data exchange between multiple nodes through the ZigBee mesh network

**Table 22:** ZigBee interface features

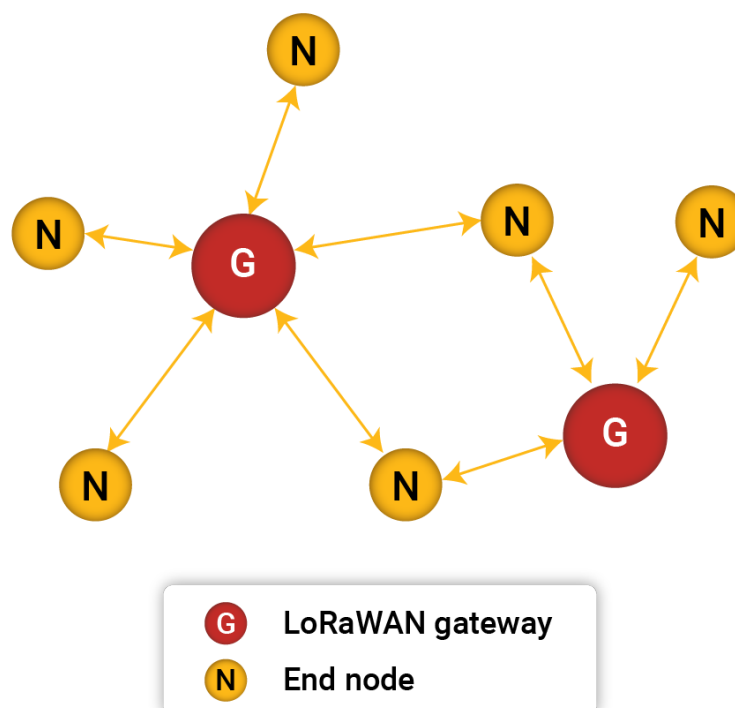
| Factors            | Details                                                          |
|--------------------|------------------------------------------------------------------|
| Type               | Low-power, low-data-rate wireless communication                  |
| Data lines         | Radio frequency (2.4 GHz ISM band, 868 MHz EU, 915 MHz US bands) |
| Communication mode | Half-duplex (CSMA/CA or mesh routing)                            |
| Number of devices  | Up to 65,000+                                                    |

| Factors       | Details                                          |
|---------------|--------------------------------------------------|
| Speed         | Up to 250 kbps                                   |
| Distance      | Up to 100 meters (with mesh networking)          |
| Applications  | Home automation, industrial control, health care |
| Advantages    | Low power, simple mesh networking                |
| Disadvantages | Low data rate, limited range                     |

## 5.4 Long range (LoRa)

As the name suggests, LoRa is a long-range, low-power, wireless communication protocol used for IoT networks, especially where wide coverage and low data rates are needed.

### LoRa wireless communication



Communication between IoT devices and a central gateway using LoRa wireless signals over sub-GHz frequency bands

LoRa end nodes use sensors to collect data and wirelessly transmit it to nearby LoRaWAN gateways, which forward it to the cloud.

Multiple gateways within range can receive the same signal, forming a star-on-star topology that enhances network reliability and coverage.

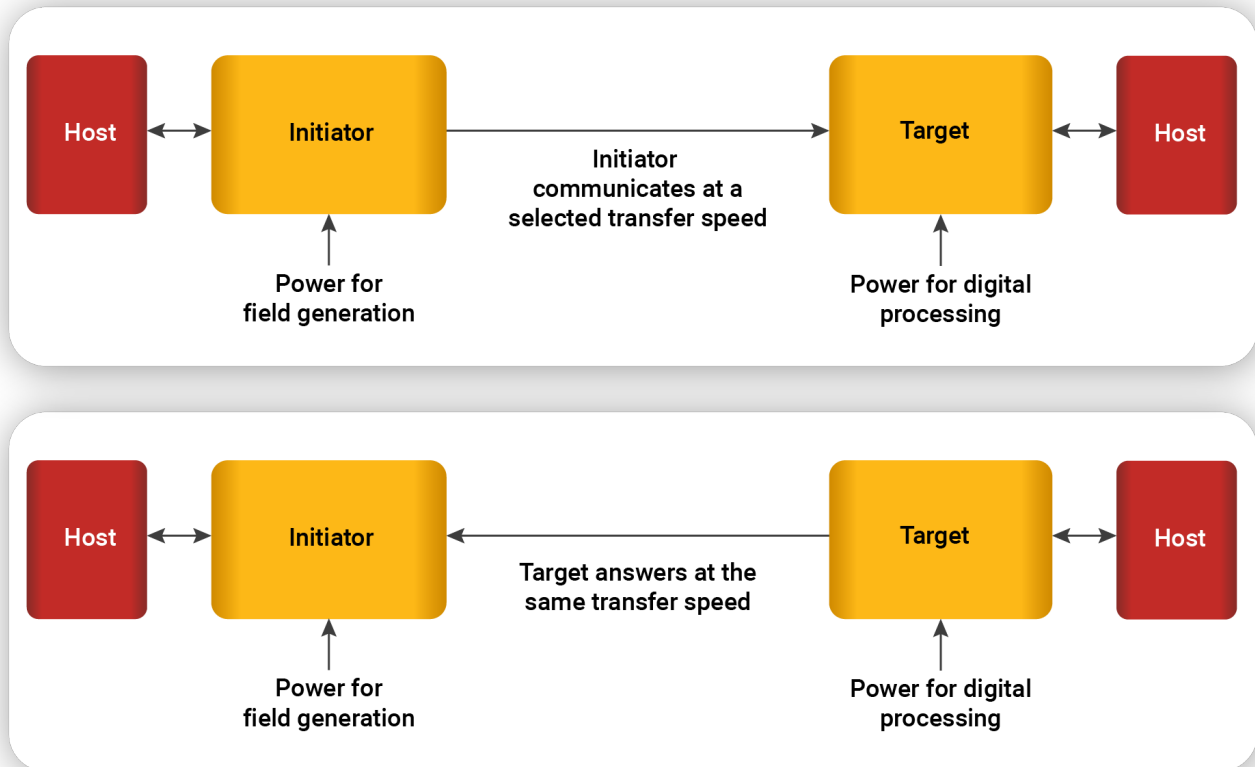
**Table 23:** LoRa interface features

| Factors            | Details                                                                    |
|--------------------|----------------------------------------------------------------------------|
| Type               | Low-power, long-range wireless communication                               |
| Data lines         | Radio frequency (sub-GHz bands)                                            |
| Communication mode | Half-duplex (LoRaWAN class-dependent)                                      |
| Number of devices  | Thousands of nodes per gateway (depends on traffic and region regulations) |
| Speed              | Up to 50 kbps                                                              |
| Distance           | Up to 15 km (in rural areas)                                               |
| Applications       | Smart cities, agriculture, industrial IoT                                  |
| Advantages         | Long range, low power consumption                                          |
| Disadvantages      | Low data rates, requires dedicated gateways                                |

## 5.5 Near field communication (NFC)

NFC is a short-range wireless communication protocol enabling data exchange over distances of a few centimeters, commonly used for contactless payments.

## NFC short-range pairing



NFC communication between the initiator and target at a selected transfer speed

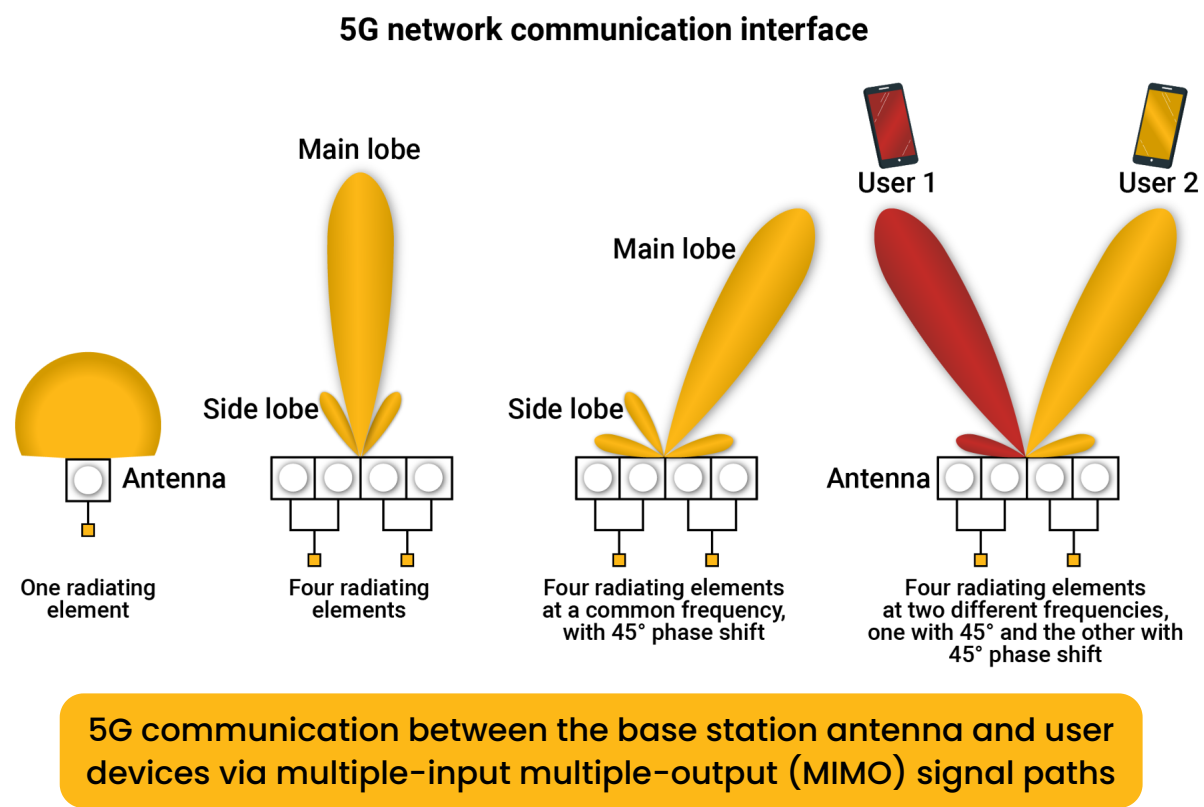
**Table 24:** NFC interface features

| Factors            | Details                                               |
|--------------------|-------------------------------------------------------|
| Type               | Short-range wireless communication                    |
| Data lines         | Radio frequency (13.56 MHz)                           |
| Communication mode | Half-duplex (alternating communication)               |
| Number of devices  | 2                                                     |
| Speed              | Up to 424 kbps                                        |
| Distance           | Up to 10 cm                                           |
| Applications       | Contactless payments, access control, pairing devices |

| Factors       | Details                               |
|---------------|---------------------------------------|
| Advantages    | Very fast setup, secure communication |
| Disadvantages | Very short range, limited speed       |

## 5.6 5th generation (5G)

This fifth-generation cellular network standard offers ultra-fast data rates, low latency, and massive device connectivity for mobile and IoT applications.



**Table 25:** 5G interface features

| Factors            | Details                                                          |
|--------------------|------------------------------------------------------------------|
| Type               | Next-generation mobile network                                   |
| Data lines         | Radio frequency (sub-6 GHz and mmWave bands)                     |
| Communication mode | Full-duplex (TDD/FDD depending on band)                          |
| Number of devices  | Up to 1 million per sq. km (theoretical in dense IoT deployment) |

| Factors       | Details                                             |
|---------------|-----------------------------------------------------|
| Speed         | Up to 10 Gbps                                       |
| Distance      | Shorter ranges for mmWave, longer for Sub-6 GHz     |
| Applications  | Smartphones, IoT, smart cities, autonomous vehicles |
| Advantages    | Ultra-high speed, low latency, longer distance      |
| Disadvantages | High infrastructure cost, limited rural coverage    |

### 5.7 Satellite communication

This wireless technology enables long-distance communication using satellites to transmit and receive signals between ground stations and devices globally.

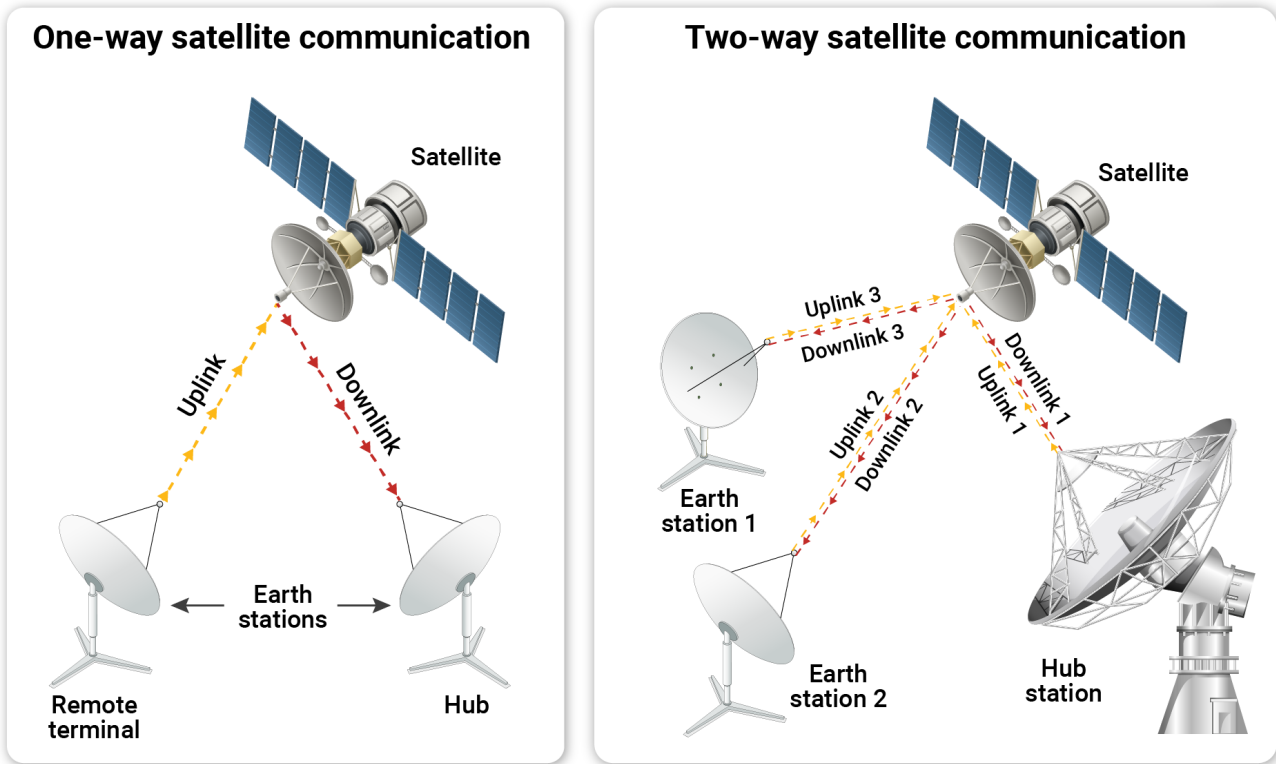
The IEEE categorizes satellite communication frequencies into the C, X, Ku, and Ka bands, as outlined below.

**Table 26:** IEEE categorization of satellite communication frequencies

| Band | IEEE designation | Uplink (GHz) | Downlink (GHz)            | Applications                                                      |
|------|------------------|--------------|---------------------------|-------------------------------------------------------------------|
| C    | 4–8 GHz          | 5.8–6.725    | 3.4–4.2                   | Fixed satellite services (FSS), TV broadcast, enterprise networks |
| X    | 8–12 GHz         | 7.9–8.4      | 7.25–7.75                 | Military and government communications                            |
| Ku   | 12–18 GHz        | 14.0–14.5    | 10.7–12.75                | VSAT, direct-to-home TV (DTH), commercial broadcasting            |
| Ka   | 26.5–40 GHz      | 27.5–31.0    | 18.3–18.8 or 19.7 to 20.2 | High-throughput satellites (HTS), broadband internet              |



The exact frequency ranges can vary by region and application, and are often classified to prevent interference or ensure security.



Satellite communication between Earth stations via uplink and downlink RF signals

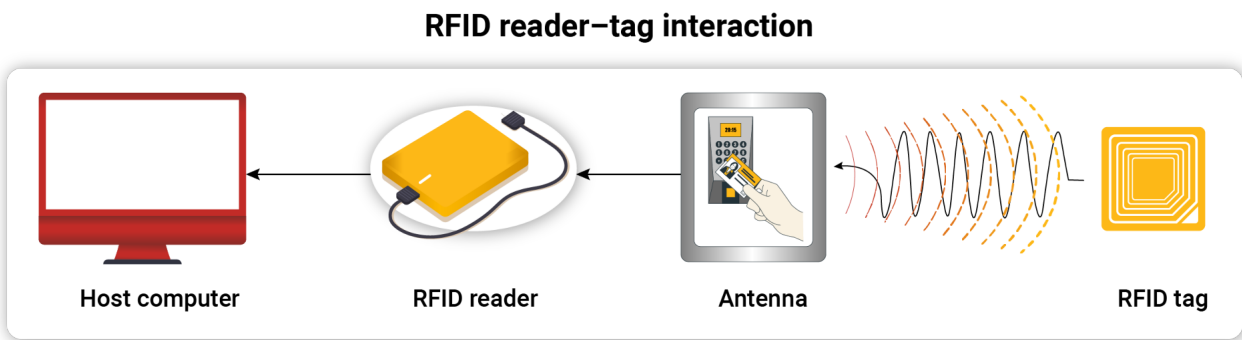
Table 27: Satellite communication interface features

| Factors            | Details                                             |
|--------------------|-----------------------------------------------------|
| Type               | Long-distance wireless communication via satellites |
| Data lines         | Radio frequency (multiple bands: C, Ku, Ka)         |
| Communication mode | Full-duplex or Half-duplex (depends on the system)  |
| Number of devices  | Thousands to millions                               |
| Speed              | Up to 100 Mbps or higher (modern systems)           |
| Distance           | Thousands of kilometers                             |

| Factors       | Details                                            |
|---------------|----------------------------------------------------|
| Applications  | Remote internet, navigation (GPS), TV broadcasting |
| Advantages    | Global coverage, reliable in remote areas          |
| Disadvantages | High latency, expensive equipment                  |

### 5.8 Radio frequency identification (RFID)

It is a wireless identification system that uses electromagnetic fields to read and write data from tags attached to objects, often used in tracking and access control.



**RFID communication between the reader and tag via electromagnetic waves in the radio frequency spectrum**

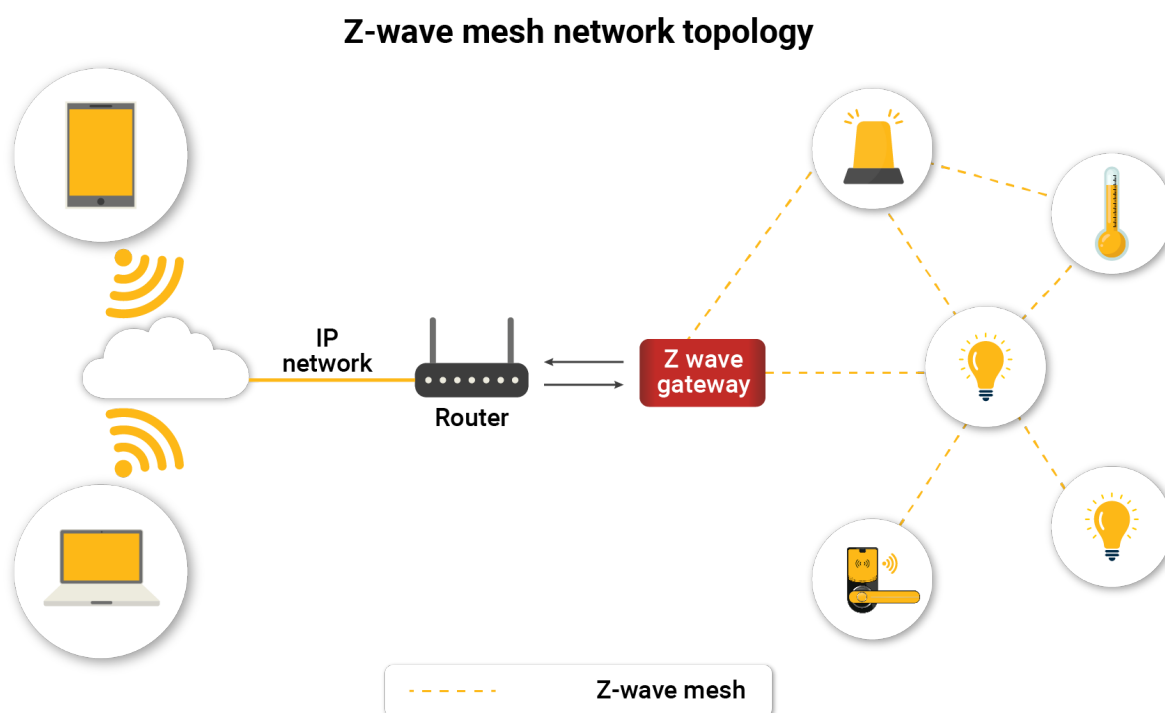
**Table 28:** RFID interface features

| Factors            | Details                                                       |
|--------------------|---------------------------------------------------------------|
| Type               | Short-range identification and tracking                       |
| Data lines         | Radio frequency (125 kHz, 13.56 MHz, UHF)                     |
| Communication mode | Half-duplex (reader initiates, tag responds)                  |
| Number of devices  | Hundreds to thousands (depending on reader type and tag type) |
| Data rate          | Depends on the system; typically low to moderate              |

| Factors       | Details                                           |
|---------------|---------------------------------------------------|
| Distance      | Few centimeters to several meters (UHF RFID)      |
| Applications  | Inventory tracking, access control, and logistics |
| Advantages    | No line-of-sight needed, fast scanning            |
| Disadvantages | Susceptible to interference, security risks       |

## 5.9 Z-Wave

Z-Wave is a low-power, wireless communication protocol designed for smart home and automation systems, commonly integrated into PCBs to enable device connectivity. It operates on a mesh network topology, allowing devices to relay messages to extend range and reliability.



**Z-Wave communication between smart home devices using low-power RF signals in the sub-GHz band**

**Table 29: Z-Wave interface features**

| Factors            | Details                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------|
| Type               | Wireless mesh network communication interface                                                      |
| Data lines         | Radio frequency (sub-GHz bands: 868.42 MHz EU, 908.42 MHz US)                                      |
| Communication mode | Half-duplex (bidirectional, but not simultaneous)                                                  |
| Number of devices  | Supports up to 232 nodes in a mesh network                                                         |
| Speed              | 9.6 kbps to 100 kbps, depending on the Z-Wave version                                              |
| Clocking           | Asynchronous, based on internal device timing                                                      |
| Distance           | 30–100 meters indoors, longer with mesh relay between nodes                                        |
| Applications       | Smart home devices, lighting control, door locks, thermostats, sensors                             |
| Advantages         | Low power, interoperable, robust mesh networking, operates in the sub-GHz band (less interference) |
| Disadvantages      | Lower data rate, limited bandwidth, requires Z-Wave-certified hardware                             |

**Key takeaways:**

1. A PCB interface defines the methods for data exchange over physical connections.
2. Connectors are the physical hardware links, while printed circuit board interfaces define the communication protocols that operate over those connections.
3. Common wired interfaces include UART, SPI, I2C, I2S, RS-232, RS-485, USB, Ethernet, FireWire, CAN, LVDS, and PCIe.
4. Key wireless interfaces comprise Bluetooth, Wi-Fi, LoRa, ZigBee, and NFC, used for cable-free communication between devices.
5. Each interface has unique properties: data lines, communication mode, speed, distance, applications, advantages, and disadvantages.

6. Understanding interface specifics helps PCB designers optimize board layout, enhance signal integrity, ensure electrical and protocol compatibility between components, and improve overall system reliability.

Understanding PCB interfaces is essential for designing robust electronic systems. By selecting and implementing appropriate interfaces, electrical engineers can ensure low-loss communication between components in modern devices.

# About Sierra Circuits



Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

## Talk to our experts

Sierra Circuits helps PCB designers plan it right! Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view. Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Why Sierra Circuits?

**Customer-oriented approach:** We collaborate closely with you and tailor solutions to meet your unique design requirements.

**Rapid high-tech innovation:** We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

**Integrated systems:** Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages at the beginning of all of our partnerships.

**Focus ready:** We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits’ PCB manufacturing capabilities overview

Rigid PCBs

| PCB highlights                       | Fabrication highlights | Quality management systems |
|--------------------------------------|------------------------|----------------------------|
| Up to 30 layers                      | Routed arrays          | AS9100D                    |
| Board thickness: .005" - .250"       | Edge plating           | ISO 9001:2015              |
| Max panel size: 21" x 29"            | Bevels                 | ISO 13485:2016             |
| Min trace and space: .002"           | Heat sinks             |                            |
| Solder mask feature tolerance: .001" |                        |                            |
| Blind and buried vias                |                        |                            |

HDI and microelectronics

| High tech in small form                | Features                                               | Certified for your industry |
|----------------------------------------|--------------------------------------------------------|-----------------------------|
| Special stack-up help / NPI review     | Down to 1.5-mil trace space, 2-mil holes               | Mil-spec                    |
| Up to 30 layers                        | Blind vias, buried vias, and other microvia techniques | Automotive                  |
| Files evaluated by engineers           | Up to 5 sequential lamination cycles                   | Aerospace and defense       |
| Quote sent by email within a few hours | Laser direct imaging                                   | Medical devices             |
|                                        | via-in-pad technology                                  |                             |

## Flex and rigid-flex PCBs

| Surface finishes           | Standard flex materials                        | Fabrication highlights | PCB classification          |
|----------------------------|------------------------------------------------|------------------------|-----------------------------|
| HASL                       | 0.5 mil to 5 mil thick polyimide material      | Routed array           | IPC 6012, Class 1, 2, and 3 |
| Lead-free HASL             |                                                | V score                |                             |
| OSP (Shikoku F2 and Entek) | 1 mil to 5 mil thick copper clad base material | Edge plating           | ISO:9001:2015               |
| ENIG                       | Flame retardant laminate                       | Edge castellation      | ISO 13485: 2015             |
| Immersion silver           |                                                | Countersink            |                             |
| Immersion white tin        | UL and RoHS compliant                          | Bevel                  |                             |
| Tin nickel                 |                                                | Milling                |                             |
| Selective gold             |                                                |                        |                             |

## Aerospace and defense PCBs

| Capabilities                             | Documents we provide                                 | Ideal for                                                               |
|------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------|
| Layer count: Up to 30                    | Certificate of conformance                           | DOD product designers developing systems for mission critical functions |
| Minimum board thickness:10 mil           | Material specifications                              |                                                                         |
| Thickness tolerance: 10%                 | Reflow profile copy (included with first article)    | Contract manufacturers who need fast, predictable lead times            |
| Minimum trace and space width: 3 mil     | Photo requirements                                   |                                                                         |
| Minimum drill-to-copper clearance: 6 mil | First article inspection report                      | Purchasing teams who require pricing and supply-chain transparency      |
| Maximum panel size: 21" x 29"            | Record of calibrated tools used during manufacturing |                                                                         |
| Controlled impedance                     | FPT, AOI, ionic cleanliness report                   | Aerospace engineers looking to shore-up their supply chain troubles     |
| Stacked vias                             |                                                      |                                                                         |



RF and microwave PCBs

| Capabilities                                                                                  | Certifications                                     |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------|
| Operating frequency:<br>10 MHz to 30 GHz                                                      | PC Manufacturers<br>Qualification Profile<br>(MQP) |
| Sodium etch hole<br>treatment process                                                         | IPC-A-600                                          |
| Full-body gold plating<br>up to 120u"                                                         | IPC-6012 standards                                 |
| Soft gold without<br>nickel or ENEPIG<br>surface finishes for<br>enhanced board<br>protection | ISO 9001:2015                                      |
|                                                                                               | NADCAP                                             |
|                                                                                               | ITAR registered                                    |
| Design assistance                                                                             | RoHS/REACH<br>standards                            |

Sierra Circuits’ PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

| Full turnkey PCB       | Partially consigned assembly        | Consigned assembly                                                  | Assembly capabilities                                                               |
|------------------------|-------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Design review          | Some parts provided by the designer | Components are provided by the PCB designer                         | BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components |
| Bare board fabrication | We will source the rest             | Sierra Circuits will fully assemble your PCBs and ship them to you. | DFA                                                                                 |
| Component procurement  | Supply chain resilient              | Quick turn times                                                    | RoHS, leaded, indium, clean and no clean chemistry                                  |
| Testing                | Quick turnaround time               | Fully transparent pricing                                           | Component sizes: 0201, 01005, 08004                                                 |
| Quick turnaround time  |                                     |                                                                     | Paste in-hole                                                                       |

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

## **Sierra Circuits' PCB design services**

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of  $\pm 5\%$
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

## **Advanced PCB design tools**

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

## **Sierra Circuits PCB certification and registration**

### **IPC certified for quality standards in PCB fabrication and assembly**

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

### **ISO certified for military, aerospace, and medical device PCB fabrication and assembly**

- ISO 9001:2015 certificate
- AS9100D – military and aerospace
- ISO 13485:2016 certificate – medical devices

### **Mil-Spec certified and DLA approved**

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

### **ITAR Registered**

- ITAR registration letter

### **Underwriters Laboratory certified**

- Flex and rigid flex PCBs
- Rigid PCBs

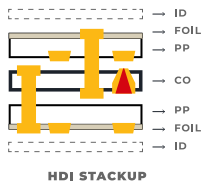
### **Exceptional material compliance and reporting**

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

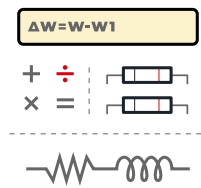
### **Certified minority supplier by NMSDC**

- Minority business certification

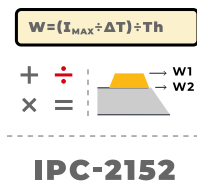
# The Designer's Toolkit



Stackup Designer

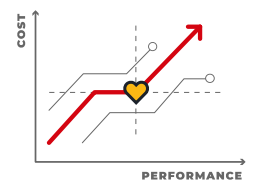


Impedance Calculator

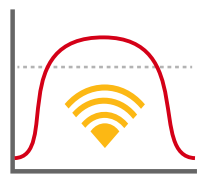


IPC-2152

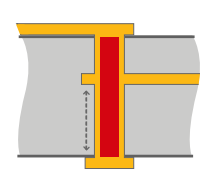
Trace Width,  
Current Capacity and  
Temperature Rise  
Calculator



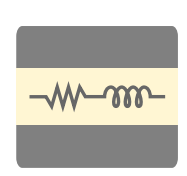
Material Selector



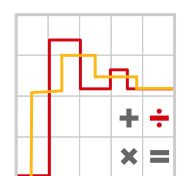
Bandwidth, Rise Time  
and Critical Length  
Calculator



Maximum Via  
Stub Length  
Calculator



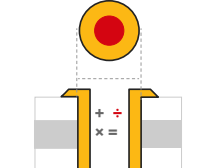
Signal Layer Estimator



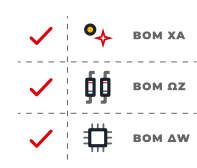
Transmission Line  
Reflection Calculator



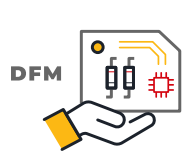
RLC Resonant  
Frequency &  
Impedance Calculator



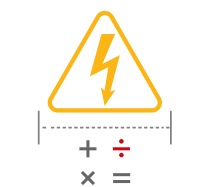
Via Current  
Capacity and  
Temperature Rise  
Calculator



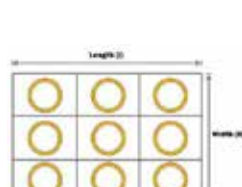
BOM Checker



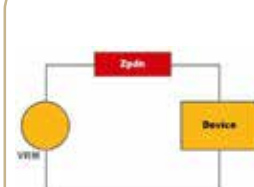
Better DFM



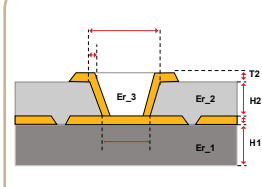
Conductor Spacing  
and Voltage Calculator



Via Thermal  
Resistance Calculator



Power Distribution  
Network Analyzer



Via Impedance  
Calculator

Try Tools

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